JC371 Communication Protocol and Data Format

Version: V1.5.2

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Revision History

Version	Date	Editor(s)	Change Location	Remarks
V1.0	Dec. 8. 2020	Tan Senwen	Entire document	Created the initial draft.
V1.1	Jun. 2, 2023	Guo Jie	4.2.7	Added the Jimi IoT specific extension to the pass-through protocol, so all device parameters can be queried and modified using pass-through commands.
V1.1	Aug. 17, 2023	Guo Jie	4.2.7	Added the capability to actively synchronize parameter changes to the platform after modifications were made via SMS, WiFiKit, BLE, or PC Tools.
V1.1	Jan. 12, 2024	Guo Jie	4.2.7	Added the specific format for the pass-through command 0xF0.
V1.2	Sept. 5, 2024	Zhang Chuangjun	Chapter 7	Added descriptions for the standard alarm and the E8 extended alarm.
V1.3	Dec. 17, 2024	Guo Jie	Chapters 3, 4, 6, and 7	Added a description of the encryption method and the packet segmentation. Added two new tables to Chapter 4: Definitions of Alarm Flags and Definitions of Status Bits, providing comprehensive details on the Alarm flag and Status fields in 0x0200 message. Added a new table to Chapter 6: Definitions of Video Alarm Flags, clarifying the corresponding video alarm flags. Enhanced Chapter 7 with a table detailing supplementary information IDs and with descriptions for the custom extension of positioning data and timezone information.
V1.4	Jan. 10, 2025	Guo Jie	Appendix	Added an appendix about data examples for uplink and downlink packets
V1.4.1	Jun. 26, 2025	Guo Jie, King Wong	Section 7.2.1	Added video alarm ID 0x000C in E8.
V1.4.2	Aug. 15, 2025	Guo Jie, King Wong	Section 7.2.3	Added basic alarm ID 0x0C1A in E8.
V1.4.3	Sep. 01, 2025	King Wong	Sections 7.2 and 7.3	1. Added section 7.3.5. 2. Added threes device status IDs –

Version	Date	Editor(s)	Change Location	Remarks
				0x0005, 0x0006, and 0x0007 – to E8 0x2005. 3. Updated the E8 Extended Alarm Data Structure Table in Section 7.2.
V1.5.0	Oct 28, 2025	King Wong	Chapters 8 and 9	1. Added new Chapter 8. (The content of the previous Chapter 8 has been moved.) 2. Moved the content of the previous Chapter 8 to Chapter 9.
V1.5.1	Nov 04, 2025	King Wong	Chapter 6	Added Section 6.2, and the numbering of all subsequent sections was automatically incremented.
V1.5.2	Feb 24, 2026	King Wong	Sections 7.1.1 & 7.1.2	Updated the data type for longitude and latitude from DWORD to INT32.

Contents

1. Terms and Abbreviations

Terms	Abbreviation/Definition
Abnormal data communication link	The radio communication link is interrupted (such as poor signal) or temporarily suspended (such as during a call or device offline).
Location reporting strategy	Timed reporting
Location reporting program	Rules for determining the interval of periodic reporting based on relevant conditions.
Additional points report while turning	The terminal sends location information when it detects the vehicle is turning. The sampling frequency is no less than 1 Hz, the vehicle's heading change rate is at least 30 °/s, and this condition lasts for at least 2 seconds.

2. Relevant Description

2.1 Notes on Versions

This document supports JT/T 808-2019 (v2019), JT/T 808-2011 (v2011), and JT/T 1078-2016. The primary focus is on v2019, while compatibility with v2011 is maintained, and JT/T 1078-2016 serves as a supplementary.

The main differences among v2011 and v2019 mentioned in this document lie in the **message header structure**, **terminal registration protocol**, **authentication protocol**, **text information delivery** (0x8300), and **packet segment retransmission request**. Chapters 4 through 6 define and describe the message body. If the content is the same in both v2011 and v2019 protocols, it will be presented in a single table. Each table title will indicate its location in the corresponding protocol, so special attention should be paid to these distinctions.

2.2 Protocol Extension

Based on the standard protocol framework issued by the Ministry of Transport (PRC) and the functional needs of current products, new functional protocols have been added to enable interaction between the platform and the terminal.

2.3 Notes on Padding

Since the v2011 and v2019 protocols use different padding methods (zero padding at the front or end), this document standardizes it by using end-padding for all fields except for phone numbers in the message header, where front-padding is retained.

The known fields requiring padding include:

(1) In the message headers of both v2011 and v2019, phone numbers are padded with zeros at the front, and this remains unchanged in this document.

(2) In terminal registration (0x0100), the terminal model and terminal ID in v2011 are end-paded, while it is front-padded in v2019. Since the original text does not specify the padding method for terminal ID, both versions will use end-padding for the terminal ID in this document.

(3) In v2019, the software version number in terminal authentication uses end-padding, and this remains unchanged in this document.

3. Protocol Format

3.1 Data Type

Table 3-1 Data Types

Data Type	Description/Requirements
BYTE	Unsigned single-byte integer (8 bits)
WORD	Unsigned double-byte integer (byte, 16 bits)
DWORD	Quad-byte unsigned integer (32 bits)
BYTE[n]	n bytes of data
BCD[n]	8421 code, n bytes
BYTE[21]	GBK encoding. Empty if no data is present.

3.2 Message Structure

3.2.1 General Structure

Each message is consisted of IDs, a message header, a message body, and an error check code. The message structure is shown as follows:

Table 3-2 Message Structure

Flag (0x7E)	Message Header	Message Body	Error Check Code	Flag (0x7E)

JT/T 808-2011 Message Structure

The message structure specified in the Ministry's standard protocol is as follows:						
		Flag (0x7E)				Flag (0x7E)
		0x7E → 0x7D 0x02				0x7E → 0x7D 0x02
		0x7D → 0x7D 0x01				0x7D → 0x7D 0x01
		Flag (0x7E)	Flag (0x7E)	Flag (0x7E)	Flag (0x7E)	Flag (0x7E)
					BYTE	
					From the message header onward, XOR with the next byte until the byte before the checksum, occupying one byte.	
	Message ID	Message Body Attributes	Terminal Phone Number	Message Sequence Number	Message Envelope Items	
Byte Position	0–1	2–3	4–9	10–11		
	WORD	WORD	BCD			
					Total Packet Count	Packet Sequence Number
				Byte Position	0–1	2–3
					WORD	WORD
	Reserved	Segmentation	Data Encryption Method	Message Body Length		
Byte Position	14–15	13	10–12	0–9		
				Custom Content		
				Refer to specific communication functions for details.		

3.2.2 Flag

The flag should be represented by 0x7E. If either 0x7E or 0x7D appears in the checksum, message header, or message body, escape processing must be applied.

Escape rules are defined as follows:

- First escape 0x7D to a fixed two-byte data: 0x7D 0x01; then escape 0x7E to a fixed two-byte data: 0x7D 0x02.

The escape process is as follows:

- When sending a message: Encapsulate the message first, then calculate and fill in the error check code, and finally perform escaping.
- When receiving a message: Perform escape restoration, then verify the checksum, and finally parse the message.

Example: The original packet 0x30 0x7e 0x08 0x7d 0x55 will be encapsulated as 0x7e 0x30 0x7d 0x02 0x08 0x7d 0x01 0x55 0x7e after escaping.

3.2.3 Message Header

Table 3-3 Message Header Structure

Start Byte		**Field**	**Data Type**		**Description/Requirements**
v2011	v2019		v2011	v2019	
0	0	Message ID	WORD		—
2	2	Message body attributes	WORD		See the table below for the structure.
—	4	Protocol version number (no such field in v2011)	BYTE		Indicate the protocol version, which is 1 for the initial version and increments with each critical revision.
4	5	Device IMEI	BYTE[6]	BYTE[10]	Convert the hex to decimal to get the first 14 digits of the device IMEI. For example: Device IMEI (hex): 4EB6FB4AD5FB Decimal equivalent: 86547807000059 Get the check digit "3" by Luhn algorithm Then the IMEI is "865478070000593" If the length of the number doesn't meet requirements, 0s are prepended; for Hong Kong, Macau, and Taiwan, the padding is based on their area codes.
10	15	Message sequence number	WORD		Send in order, starting from 0 and adding cyclically
12	17	Message encapsulation item	—		This field has content if the relevant flag in message body attributes field confirms segmentation; otherwise, no such field exists.

Example data: 7E 02 00 00 5F 4E B6 FB 4A D5 FB 00 04 00 00 00 00 00 0C 00 03 01 58 7D B2 06 CA A2 48 00 3B 00 00 00 00 25 01 16 16 17 45 01 04 00 00 00 02 14 04 00 00 00 01

30 01 1F 31 01 1B 15 04 00 00 00 02 E8 1E 04 11 10 30 30 30 30 35 39 33 25 01 16 16 17 45 00 00 00 00 20 03 07 01 00 00 00 00 06 42 EB 09 55 54 43 2B 30 38 3A 30 30 AE 7E.

Conversion:

- Device ID (Hex): 4EB6FB4AD5FB
- Decimal equivalent: 86547807000059
- Then apply the Luhn algorithm to calculate the 15th check digit and get "3".
- Full IMEI (decimal): 865478070000593

Table 3-4 Structure of Message Body Attributes

<table>
<tr><th>BIT</th><th>15</th><th>14</th><th>13</th><th>12</th><th>11</th><th>10</th><th>9</th><th>8</th><th>7</th><th>6</th><th>5</th><th>4</th><th>3</th><th>2</th><th>1</th><th>0</th></tr>
<tr><td>v2011</td><td colspan="2">Reserved for future use</td><td>Segmentation</td><td colspan="3">Data encryption method</td><td colspan="10">Message body length</td></tr>
<tr><td rowspan="2">v2019</td><td>Reserved for future use</td><td>Version identifier</td><td>Segmentation</td><td colspan="3">Data encryption method</td><td colspan="10">Message body length</td></tr>
<tr><td colspan="16">NOTE: The version identifier's value is fixed as 1.</td></tr>
</table>

Notes on data encryption method:

- Bits 10 to 12 are data encryption flag bits;
- If the three bits are 0, the message body is unencrypted (default);
- If bit 10 is 1, the message body is encrypted using RSA;
- The remaining bits are reserved.

Packet segmentation:

If bit 13 in the message body attributes is 1, the message body is long and requires segmentation prior to transmission. The segmentation details are determined by the message encapsulation item. If bit 13 is 0, there is no message encapsulation item field in the header.

The content of the message encapsulation item is shown in the table below:

Table 3-5 Message Encapsulation Item Content

Start Byte	Field	Data Type	Description/Requirements
0	Total packet count	WORD	Total number of packets after segmentation
2	Packet sequence number	WORD	Start from 1.

3.2.4 Message Body

Different message bodies correspond to specific message IDs; their content and formats can be referred to in Chapters 4 to 6.

3.2.5 Error Check Code

The checksum is calculated starting from the **very first byte of the message header**. XOR operations are performed with each subsequent byte until the **last byte of the message body**. The result of these operations is a one-byte checksum.

4. General Protocols (Message Body)

The protocols are classified and described by functionality. Unless otherwise specified, TCP is used by default.

4.1 Basic Protocols

4.1.1 General Terminal Response (0x0001)

Table 4-1 Format of General Terminal Response Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Reply sequence number	WORD	Correspond to the sequence number of the platform-initiated message.
2	Response ID	WORD	Correspond to the ID of the platform-initiated message.
4	Result	BYTE	0: Success/Confirmation; 1: Failure; 2: Message error; 3: Not supported.

4.1.2 General Platform Response (0x8001)

Table 4-2 Format of General Platform Response Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Reply sequence number	WORD	Correspond to the sequence number of the terminal-initiated message.
2	Response ID	WORD	Correspond to the ID of the terminal-initiated message.
4	Result	BYTE	0: Success/Confirmation; 1: Failure; 2: Message error; 3: Not supported; 4: Alarm processing confirmation.

4.1.3 Terminal Heartbeat (0x0002)

The heartbeat message body is empty.

4.1.4 Terminal Registration (0x0100 / 0x8100)

The terminal must first register. Upon registration success, the terminal obtains and saves the authentication code, which will be needed for future authentication. Before terminal removal or replacement, the terminal must be deregistered to unbind from the vehicle.

If SMS is used send terminal registration and deregistration messages, the platform should reply to the registration message with an SMS response and to the deregistration message with a general response (0x8001) via SMS.

- **Terminal registration message ID: 0x0100**

See the table below for the registration message body format.

Table 4-3 Registration Message Body Format

Start Byte		**Field**	**Data Type**		**Description/Requirements**	
v2011	v2019		v2011	v2019	v2011	v2019
0		Province ID	WORD		Indicate the province where the vehicle installed with the terminal is. The platform will assign a default value if this field is set to 0. The ID is the first two digits of the administrative code specified in GB/T 2260 *Codes for the administrative divisions of the People's Republic of China*.	
2		City/County ID	WORD		Indicate the city and county where the vehicle installed with the terminal is. The platform will assign a default value if this field is set to 0. The ID is the last four digits of the administrative code specified in GB/T 2260 *Codes for the administrative divisions of the People's Republic of China*.	
4		Manufacturer ID	BYTE[5]	BYTE[11]	Consist of the administrative code (of the manufacturer's location) and manufacturer ID.	
9	15	Device model	BYTE[20]	BYTE[30]	Specify by the manufacture and occupy 20 bytes, where "0x00" are appended if the length doesn't meet the requirement.	Specify by the manufacture and append with "0x00" if the length doesn't meet the requirement.
29	45	Terminal ID	BYTE[7]	BYTE[30]	Specify by the manufacture and occupy 7 bytes,	The terminal ID is specified by the manufacturer and

Start Byte		Field	Data Type		Description/Requirements	
v2011	v2019		v2011	v2019	v2011	v2019
					where the ID should be a combination of upper-case letters and digits and "0x00" are appended if the length doesn't meet the requirement.	should be a combination of upper-case letters and digits.
36	75	License plate color	BYTE		Follow what is specified in JT/T 697.7-2014 *Basic data element of transportation information* and set it to 0 for unregistered vehicles.	
37	76	License plate number	BYTE[21]		Issued by the Traffic Management Department under Public Security Bureau. Input the VIN for unregistered vehicles.	

Note:

The Terminal ID is the IMEI of the terminal (for example: 323433313137309).

The Terminal ID in v2011 is 7-byte long (BYTE[7]): 0x32 0x34 0x33 0x31 0x31 0x37 0x30 (discard the last bit).

The Terminal ID in v2019 is 30-byte long (BYTE[30]):

0x32 0x34 0x33 0x31 0x31 0x37 0x30 0x90 0x00 0x00

0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00

0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 (appended with 0s)

- **Terminal registration response message ID: 0x8100**

See the table below for the registration response message body format.

Table 4-4 Registration Response Message Body Format

Start Byte	Field	Data Type	Description/Requirements
0	Reply sequence number	WORD	Correspond to the sequence number of the terminal registration message.
2	Result	BYTE	0: Success; 1: Vehicle already registered; 2: No such vehicle in database; 3: Terminal already registered; 4: No such terminal in database

Start Byte	Field	Data Type	Description/Requirements
3	Authentication code	BYTE[21]	This field exists only when the registration result is 0.

4.1.5 Terminal Deregistration (0x0003)

The terminal parameter query message body is empty.

4.1.6 Terminal Authentication (0x0102)

Upon successful connection to the platform, registered terminals must immediately authenticate. No other messages should be transmitted until authentication is successful.

The authentication process involves the terminal sending an authentication message, to which the platform replies with a general response message (0x8001).

- **v2011:**

Table 4-5 Authentication Message Body Format

Start Byte	Field	Data Type	Description/Requirements
0	Authentication code	BYTE[21]	The terminal sends this code to the platform upon reconnection.

- **v2019**

Table 4-6 Authentication Message Body Format

Start Byte	Field	Data Type	Description/Requirements
0	Authentication code length	BYTE	—
n	Authentication code content	BYTE[21]	Length of authentication code (n)
n+1	Terminal IMEI	BYTE[15]	—
n+9	Software version number	BYTE[20]	Specify by the manufacturer and append with 0x00 if the length doesn't meet requirements. 'n' represents the length of the authentication code.

4.1.7 Segment Retransmission Request (0x8003 / 0x0005)

- Message ID for server-initiated segment retransmission requests (v2019): 0x8003
- Message ID for terminal-initiated segment retransmission requests (v2019): 0x0005

The format of the terminal-initiated segment retransmission request message body is the same as that of the server-initiated segment retransmission request message body, as shown in the table below.

NOTE: The v2011 has only 0x8003.

Table 4-7 Format of Segment Retransmission Request Message Body

Start Byte		Field	Data Type		Description/Requirements
v2011	v2019		v2011	v2019	
0		Original message sequence number	WORD		Correspond to the sequence number of the first segment to be retransmitted.
4	2	Retransmission segment count	BYTE	WORD	n
5	4	Retransmission segment ID list	BYTE[2*n]		Ordered by the retransmission segment sequence number, such as Segment ID1, Segment ID2, ..., Segment IDn. "n" is the total number of retransmission segments.
NOTE: The response to this message should be to retransmit the segment(s) in the retransmission segment ID list exactly the same as the original segment message.					

4.2 Terminal Management Protocols

Parameters are typically used for configuring various functions of a terminal. Each parameter has a unique ID and data type, as detailed in the table below. This table applies to protocols 0x8103, 0x8106, and 0x0104.

Table 4-8 Definitions and Descriptions of Parameter IDs

Parameter ID	Data Type	Description/Requirements
0x0001	DWORD	Interval to send heartbeat packets (unit: second)
0x0010	BYTE[21]	Primary server access point name (APN) for wireless communication.

Parameter ID	Data Type	Description/Requirements
		If the network type is CDMA, this will be the PPP dial-up number.
0x0011	BYTE[21]	Primary server dial-up username for wireless communication
0x0012	BYTE[21]	Primary server dial-up password for wireless communication
0x0013	BYTE[21]	Primary server address - IP or domain name, where the host and port number are separated by a colon (:) and multiple servers are separated by semicolons (;).
0x0018	DWORD	TCP port
0x0020	DWORD	Location reporting strategy (0: Timed reporting; 1: Distance-based reporting; 2: both)
0x0027	DWORD	Reporting interval in sleep mode, in seconds (s). The value must be greater than 0.
0x0029	DWORD	Default reporting time interval in seconds (s). The value must be greater than 0.
0x002C	DWORD	Default reporting distance in meters (m). The value must be greater than 0.
0x0030	DWORD	Turning angle (degrees) to trigger additional location data upload. The value must be smaller than 180.
0x0031	WORD	Geofence radius (unauthorized movement threshold) in meters (m)
0x0041	BYTE[21]	Reset phone number, used to call the terminal to reset itself.
0x0042	BYTE[21]	Factory reset phone number, used to call the terminal to restore to factory settings.
0x0045	DWORD	Terminal call answering policy (0: Answer automatically; 1: Answer automatically when ACC ON and manually when ACC OFF)
0x0048	BYTE[21]	Listen-in phone number
0x0050	DWORD	Alarm mask word, corresponding to the alarm flag in the location reporting message. If the corresponding bit is 1, the corresponding alarm will be masked.
0x0051	DWORD	Switch to enable or disable SMS alarms, corresponding to the alarm flag in the location reporting message. If the corresponding bit is 1, the corresponding alarm will be sent via SMS.
0x0052	DWORD	Switch to enable or disable camera recording for alarms, corresponding to the alarm flag in the location reporting message. If the corresponding bit is 1, the camera will be activated to record the corresponding alarm.
0x0053	DWORD	Alarm photo save flag, corresponding to the alarm flag in the location reporting message. If the corresponding bit is 1, the photo(s) taken during the corresponding alarm will be saved or upload immediately.
0x0055 (0xF008)	DWORD	Maximum speed (km/h)

Parameter ID	Data Type	Description/Requirements
0x0056	DWORD	Overspeed duration, in seconds (s)
0x0065	DWORD	Distance-based photo control
0x0080	DWORD	Vehicle odometer reading (unit: 0/10km)
0x0081	WORD	Province ID where the vehicle is located
0x0082	WORD	City ID where the vehicle is located
0x0090	BYTE	GNSS setting, defined as follows: bit0 = 0 (Disable GPS) / 1 (Enable GPS) bit1 = 0 (Disable BDS) / 1 (Enable BDS) bit2 = 0 (Disable GLONASS) / 1 (Enable GLONASS) bit3 = 0 (Disable Galileo) / 1 (Enable Galileo)
0x0075	Array	Audio/video parameter settings
0x0076	Array	Audio/video channel list
0x0077	Array	Individual video channel parameter settings
0x0079	Array	Special alarm video recording settings
0x007A	DWORD	Video-related alarm mask word, see Chapter 7 for details.

4.2.1 Terminal Parameter Setup (0x8103)

The platform sets the parameters of a terminal by sending a message carrying these parameters, and the terminal replies with a general response message (0x0001). The platform can query terminal parameters by sending a query message, and the terminal replies with the corresponding parameter settings. Terminals should support specific parameters depending on the network type.

Table 4-9 Format of Parameter Setup Request Message Body

Start Byte	Field			Data Type	Description/Requirements
0	Total number of parameters			BYTE	Number of parameter items (n)
1	Parameter item list	Parameter item 1	Parameter ID	DWORD	See Table 4-8 Definitions and Descriptions of Parameter IDs for details.
			Parameter length	BYTE	Parameter length
			Parameter value	-	If it is a multi-value parameter, multiple parameter items with the same ID are used in the message, such as the dispatch center number.
		Parameter	...	...	...

Start Byte	Field			Data Type	Description/Requirements
		item 2			
		Parameter item n	...	...	...

For example:

Set the location reporting strategy to 0 (timed-reporting), location reporting program to 0 (based on ACC status), reporting interval when ACC ON to 10s, reporting interval during SOS alarm to 60s, and reporting interval when ACC OFF (sleep) to 200s, then the message is as follows:

[v2011] 7E 81 03 00 37 01 88 12 34 56 78 E8 BE 06 00 00 00 20 04 00 00 00 00 00 00 00 21 04 00 00 00 00 00 00 00 29 04 00 00 00 0A 00 00 00 28 04 00 00 00 3C 00 00 00 27 04 00 00 00 C8 00 00 00 30 04 00 00 00 28 A5 7E

4.2.2 Terminal Parameter Query (0x8104)

The message body is empty.

4.2.3 Specified Terminal Parameter Query (0x8106)

The format of the specified terminal parameter query message body is shown below. The terminal replies with 0x0104.

Table 4-10 Format of Specified Terminal Parameter Query Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Total number of parameters	BYTE	—
1	Parameter ID list	BYTE[4*n]	Ordered by parameter sequence, such as "Parameter ID1 Parameter ID2 ... Parameter IDn", where "n" is the total number of parameters.

4.2.4 Terminal Parameter Query Response (0x0104)

Table 4-11 Format of Parameter Query Response Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Reply sequence number	WORD	Correspond to the sequence number of the

Start Byte	Field			Data Type	Description/Requirements
					terminal parameter query message.
2	Number of parameters in the reply			BYTE	n
3	Parameter item list	Parameter item 1	Parameter ID	DWORD	See Table 4-8 Definitions and Descriptions of Parameter IDs for details.
			Parameter length	BYTE	Parameter length
			Parameter value	—	See Table 4-8 Definitions and Descriptions of Parameter IDs for details.
		Parameter item 2	...	...	...
		Parameter item n	...	...	...

Example (v2011):

Platform sends:

7E 81 04 00 00 01 88 12 34 56 78 00 B4 B0 7E

Terminal replies:

7E 01 04 00 5F 01 88 12 34 56 78 00 0B 00 B4 09 00 00 00 01 04 00 00 00 78 00 00 00 10 05 63 6D 6E 65 74 00 00 00 13 0E 31 32 30 2E 32 33 37 2E 38 37 2E 31 39 34 00 00 00 18 04 00 00 EA 68 00 00 00 20 04 00 00 00 00 00 00 27 04 00 00 00 B4 00 00 00 29 04 00 00 00 0A 00 00 00 2C 04 00 00 01 2C 00 00 00 30 04 00 00 00 0F 45 7E

4.2.5 Terminal Control (0x8105)

[Application] Reset the terminal or restore the terminal to factory settings.

See the table below for the terminal control message body format.

Table 4-12 Control Message Body Format

Start Byte	Field	Data Type	Description/Requirements
0	Command Word	BYTE	See Table 4-13 Command Word Details for details.
1	Command parameter	BYTE[21]	The format of the command parameters is described below. Each field is separated by a semicolon (";"). Each STRING field is encoded using GBK before being assembled into the message.

Table 4-13 Command Word Details

Command Word	Command parameter	Description/Requirements
2	BYTE[21]	Control the terminal to connect to a specified server. Parameters are separated by semicolons (;). The control command is as follows: "Connection control (0 to switch to the designated supervision platform server, 1 to switch back to the default monitoring platform server); Supervision platform authentication code; Dial point name; Dial-up username; Dial-up password; Address; TCP port; UDP port; Connection time limit to the specified server." If a parameter has no value, leave it empty. If the connection control value is 1, no subsequent parameters are required.
4	None	Reset the terminal
5	None	Restore the terminal to default settings.

4.2.6 TTS Message Delivery (0x8300)

➢ **v2011:**

Table 4-14 Format of TTS Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Flag	BYTE	1: Emergency
1	Text message	BYTE[21]	A maximum length of 1024 bytes encoded in GBK

Example (v2011): Send the command timer,30,180#

7E 83 00 00 0E 01 88 12 34 56 78 E8 BE 01 74 69 6D 65 72 2C 33 30 2C 31 38 30 23 25 7E

➢ **v2019:**

Table 4-15 Format of TTS Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Flag	BYTE	See the table below for details.
1	Text type	BYTE	1: Notification; 2: Service
2	Text message	BYTE[21]	A maximum length of 1024 bytes encoded in GBK

Table 4-16 Connotations of Flags in TTS Message

Bit	Flag
0–1	01: Service; 10: Emergency; 11: Notification
2	1: Display on terminal screen
3	1: Read out by terminal (TTS)
4	—
5	0: Central navigation information, 1: CAN fault code information
6–7	Reserved

4.2.7 Pass-Through Protocols (0x8900 / 0x0900)

Downlink transparent message ID: 0x8900

The format of the downlink transparent message body is shown in the table below:

Table 4-17 Format of Downlink Transparent Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Transparent message type	BYTE	See Table 4-19 Definitions of Transparent Message Types for details.
1	Transparent message content	—	

Uplink transparent message ID: 0x0900

The format of the uplink transparent message body is shown in the table below:

Table 4-18 Format of Uplink Transparent Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Transparent message type	BYTE	See Table 4-19 Definitions of Transparent Message Types for details.
1	Transparent message content	—	—

Table 4-19 Definitions of Transparent Message Types

Transparent message type	Definition	Description/Requirements
Detailed GNSS location data	0x00	Detailed GNSS location data

Transparent message type	Definition	Description/Requirements
Road transport certificate (IC card) information	0x0B	The message is 64-byte long for uplink transmission and 24-byte long for downlink transmission. The timeout length of this transparent message is 30s and no retransmission will be performed after timeout.
Serial port 1 transparent transmission	0x41	Transparent message sent over serial port 1
Serial port 2 transparent transmission	0x42	Transparent message sent over serial port 2
User-defined message	0xF0–0xFF	0xF0: Online command
		0xF1: jimiiothub raw content
		0xF2: Terminal parameters, see the table below for details.

Table 4-20 Transparent Online Command

Start Byte	Field	Data Type	Description/Requirements
0	Transparent message type	0xF0	
1	Transparent message content (n)	n	Example formats: VERSION# COMVER# SOS,A,X1,X2,X3#

> **NOTE:** When the platform sends transparent commands, it uses the message sequence number to establish a one-to-one correspondence with the terminal's response. The terminal's reply will use the same sequence number as the platform's original command.

4.3 Location Information

4.3.1 Location Reporting (0x0200)

The terminal should immediately upload a location message including the alarm status to the platform when an alarm is triggered.

The location reporting message body consists of the **basic and supplementary location information**, as shown in the table below.

Table 4-21 Format of Location Reporting Message Body

/	Start Byte	Field		Data Type	Description/Requirements
Supplementary information item list	0	Alarm flag		DWORD	See Table 4-22 Definitions of Alarm Flags for details.
	4	Status		DWORD	See Table 4-23 Definitions of Status Bits for details.
	8	Latitude		DWORD	Longitude in degrees multiplied by 10^6, accurate to one millionth of a degree
	12	Longitude		DWORD	Longitude in degrees multiplied by 10^6, accurate to one millionth of a degree
	16	Elevation		WORD	Altitude in meters (m)
	18	Speed		WORD	1/10 km/h
	20	Heading		WORD	0–359 (where 0 is due north, measured clockwise)
	21	Date/time		BCD[6]	YY-MM-DD-hh-mm-ss (GMT time, which prevails in this document)
		Supplementary information item 1	Supplementary information ID	BYTE	1-255
			Supplementary information length	BYTE	x
			Supplementary information	BYTE[x]	See 7 for details.
		Supplementary information item 2	...	...	...
		Supplementary	...	...	...

/	Start Byte	Field		Data Type	Description/Requirements
		information item n			

Table 4-22 Definitions of Alarm Flags

Bit	Definition	Processing Remarks
0	1: Emergency alert triggered upon emergency button activation	Zero upon receiving a response.
1	1: Overspeed alert	Flag persists until condition cleared.
2	1: Fatigue driving	Flag persists until condition cleared.
3	1: Pre-alert	Zero upon receiving a response.
4	1: GNSS module failure	Flag persists until condition cleared.
5	1: GNSS antenna unconnected or cut	Flag persists until condition cleared.
6	1: GNSS antenna shorted	Flag persists until condition cleared.
7	1: Main power supply undervoltage	Flag persists until condition cleared.
8	1: Main power supply failure	Flag persists until condition cleared.
9	1: LCD or monitor failure	Flag persists until condition cleared.
10	1: TTS module failure	Flag persists until condition cleared.
11	1: Camera failure	Flag persists until condition cleared.
12–17	Reserved	
18	1: Excessive accumulated daily driving time	Flag persists until condition cleared.
19	1: Parking overtime	Flag persists until condition cleared.
20	1: Geofence entry/exit alert	Zero upon receiving a response.
21	1: Route entry/exit alert	Zero upon receiving a response.
22	1: Insufficient/excessive driving time in road segment	Zero upon receiving a response.
23	1: Deviate from route	Flag persists until condition cleared.
24	1: VSS fault	Flag persists until condition cleared.
25	1: Fuel level exception	Flag persists until condition cleared.
26	1: Vehicle being stolen (via anti-theft device)	Flag persists until condition cleared.
27	1: Unauthorized ignition on	Zero upon receiving a response.
28	1: Unauthorized movement	Zero upon receiving a response.
29–31	Reserved	

Table 4-23 Definitions of Status Bits

Bit	Status
0	0: ACC off; 1: ACC on
1	0: No position fix; 1: Position fix acquired
2	0: South latitude; 1: North latitude
3	0: East longitude; 1: West longitude
4	0: Operational; 1: Non-operational
5	0: Coordinates unencrypted; 1: Coordinates encrypted
6–9	Reserved
10	0: Fuel system normal; 1: Fuel system disconnected
11	0: Electrical system normal; 1: Electrical system disconnected
12	0: Door unlocked; 1: Door locked
13–31	Reserved

By default, supplementary information IDs 0x01, 0x30, and 0x31 are included in the location message. Other supplementary information is only included when required by specific functionalities. For example, if no geofence alarm is triggered, there's no need to include data with the supplementary information ID 0x12. When the ACC is ON, the data with the supplementary information ID 0x2A is unnecessary as well.

(v2011) Example: The terminal sends: 7E 02 00 00 26 01 88 12 34 56 78 00 05 00 00 00 00 00 0C 00 03 01 58 7F FE 06 CA 3B 26 00 04 00 00 00 00 17 03 07 10 51 01 2A 02 00 00 30 01 1A 31 01 15 D7 7E

The platform replies: 7E 80 01 00 05 01 88 12 34 56 78 00 02 00 05 02 00 00 00 7E

4.3.2 Location Query (0x8201)

The location query message body is empty.

(v2011) Example: The platform sends: 7E 82 01 00 00 01 88 12 34 56 78 E8 BC 56 7E

The terminal replies: 7E 02 00 00 26 01 88 12 34 56 78 00 10 Terminal replies: 00 00 00 00 00 0C 00 03 01 58 7F FE 06 CA 3B 26 00 04 00 00 00 00 17 03 07 11 13 01 2A 02 00 00 30 01 1A 31 01 13 F5 7E

4.3.3 Location Query Response (0x0201)

Table 4-24 Format of Location Query Response Message Body

Start Byte	**Field**	**Data Type**	**Description/Requirements**
0	Reply sequence number	WORD	Correspond to the sequence number of the location query message.
2	Location reporting	—	See Table 7-1 Location Reporting Message Structure for details.

4.3.4 Batch Location Data Upload (0x0704)

The format of the batch location data upload message is shown in the table below.

Table 4-25 Format of Batch Location Data Upload Message

<table>
<tr><th>Start Byte</th><th>Field</th><th>Data Type</th><th colspan="4">Remarks</th></tr>
<tr><td>0</td><td>Number of data items</td><td>WORD</td><td colspan="4">The number of location reporting data items n, which must be greater than zero.</td></tr>
<tr><td>1</td><td>Location data type</td><td>BYTE</td><td colspan="4">0: Regular location data; 2: Buffered location data (generated in dead zones)</td></tr>
<tr><td>2</td><td rowspan="3">Location reporting data item 1</td><td rowspan="3">—</td><td>Start Byte</td><td>Field</td><td>Data Type</td><td>Remarks</td></tr>
<tr><td></td><td>0</td><td>Location reporting data body length</td><td>WORD</td><td>The length of the location reporting data body is m.</td></tr>
<tr><td></td><td>2</td><td>Location reporting data body</td><td>BYTE[n]</td><td>See 4.3.1 for the definition. n is the length of the location reporting</td></tr>
</table>

Start Byte	Field	Data Type	Remarks			
						data body.
	Location reporting data item 2	—	...	...	...	...
	Location reporting data item n	—	...	...	...	...

4.4 Vehicle Control Protocols

The platform sends a vehicle control message to request the terminal to perform specified operations on the vehicle. After receiving the message, the terminal should immediately reply with a general response message (0x0001). After that, the terminal should perform the operation, and based on the result send a vehicle control response message.

4.5 Vehicle Management Protocols

The platform sends messages to set geofences (circular or rectangular or polygonal) and routes to confine the operating area(s) and traveling route(s) of the terminal, that is, the vehicle.

The terminal then determines whether the conditions for triggering alarms are met based on the attributes of these areas and routes. The alarms include overspeed, geofence entry/exit, route deviation, and insufficient or excessive driving duration on a specific road segment. The terminal should include the corresponding location-related supplementary information in the location reporting message.

The valid range for the geofence or route IDs is 1–0xFFFFFFFF. If the set ID duplicates an existing geofence or route ID of the same type in the terminal, the data for that ID will be updated.

The platform can also delete geofences (circular or rectangular or polygonal) or routes in the terminal using delete messages.

The geofence and route set or delete message requires the terminal to reply with a general response message (0x0001).

5. JT/T 808 Multimedia-Related Protocols (Message Body)

Table 5-1 Request-Response IDs Mapping Table

Event/Function	Request Message	Response Message
Trigger event	[Terminal] 0x0800 Multimedia event information upload	[Platform] 0x8001 General platform response
	[Terminal] 0x0801 Multimedia data upload	[Platform] 0x8800 Response to multimedia data upload request
Request to take photos	[Platform] 0x8801 Immediate capture command	[Terminal] 0x0001 General terminal response
		[Terminal] 0x0805 Response to immediate capture command
Retrieve directory	[Platform] 0x8802 Multimedia data retrieval	[Terminal] 0x0802 Response to multimedia data retrieval request
Request to save and upload multimedia data	[Platform] 0x8803 Stored multimedia data upload command (based on time)	[Terminal] 0x0001 General terminal response
	[Platform] 0x8805 Single stored multimedia data retrieval and upload command (based on multimedia data ID)	

5.1 Multimedia Event Information Upload (0x0800)

The format of the multimedia event information upload message is shown in the table below.

Table 5-2 Format of Multimedia Event Information Upload Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Multimedia data ID	DWORD	Value greater than 0
4	Multimedia type	BYTE	0: Photo; 1: Audio; 2: Video
5	Multimedia format	BYTE	0: JPEG; 1: TIF; 2: MP3; 3: WAV; 4: WMV; Others are reserved
6	Event item code	BYTE	0: Platform-issued command; 1: Timed action; 2: Burglary

Start Byte	Field	Data Type	Description/Requirements
			alarm; 3: Collision/rollover alarm; 4: Door open snapshot; 5: Door close snapshot; 6: Door status change (open to close), speed increases from lower than 20 km/h to higher than 20 km/h; 7: Distance-based snapshot; Others are reserved
7	Channel ID	BYTE	—

5.2 Multimedia Data Upload (0x0801 / 0x8800)

When the terminal is activated by a specific event to capture or record, it should actively upload the multimedia event information after the event occurs. This message requires the platform to reply with a general response message (0x8001).

Note (v2011): The response to this message should be to use 0x0801 to retransmit the segment(s) in the retransmission segment list exactly the same as in the original segment message.

- **Multimedia data upload message ID: 0x0801**

The format of the multimedia data upload message body is shown in the table below:

Table 5-3 Format of Multimedia Data Upload Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Multimedia data ID	DWORD	>0
4	Multimedia type	BYTE	0: Photo; 1: Audio; 2: Video
5	Multimedia format	BYTE	0: JPEG; 1: TIF; 2: MP3; 3: WAV; 4: WMV; Others are reserved
6	Event item code	BYTE	0: Platform-issued command; 1: Timed action; 2: Burglary alarm; 3: Collision/rollover alarm; Others are reserved
7	Channel ID	BYTE	
8	Location reporting (0x0200) message body	BYTE[28]	Basic location information in the multimedia data
36	Multimedia data packet		

➢ **Multimedia data upload response message ID: 0x8800**

The format of the multimedia data upload response message body is shown in the table below.

Table 5-4 Format of Multimedia Data Upload Response Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Multimedia data ID	DWORD	The value should be greater than 0. If the entire data packet has been received, there is no further fields.
4	Retransmission segment count	BYTE	
5	Retransmission segment ID list		The total number of IDs cannot be greater than 125. No such field means all segments have been received.

5.3 Immediate Capture Command (0x8801 / 0x0805)

The platform sends an immediate capture command message to the terminal, which replies with a general response message (0x0001). If immediate upload is required, the terminal immediately uploads the photo or video after taken; otherwise, to save the photo or video.

➢ **Immediate capture command message ID: 0x8801**

The format of the immediate capture command message body is shown in the table below.

Table 5-5 Format of Immediate Capture Command Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Channel ID	BYTE	Value greater than 0
1	Capture command	WORD	0: Stop capturing; 0xFFF: Record video; Other: Number of photos to be taken
3	Capture interval / recording duration	WORD	The unit is second (s). The value "0" means take photos at the minimum intervals or always keep recording.
5	Save flag	BYTE	1: Save; 0: Upload now
6	Resolution	BYTE	0x00: Lowest resolution; 0x01: 320 x 240; 0x02: 640 x 480; 0x03: 800 x 600; 0x04: 1024 x 768; 0x05: 176 x 144 (Qcif);

Start Byte	Field	Data Type	Description/Requirements
			0x06: 352 x 288 (Gif); 0x07: 704 x 288 (Half D1); 0x08: 704 x 576 (D1); 0xff: Highest resolution
7	Photo / video quality	BYTE	Range from 1 to 10, where 1 represents the minimum quality loss and 10 the highest compression.
8	Brightness	BYTE	0-255
9	Contrast	BYTE	0–127
10	Saturation	BYTE	0–127
11	Hue	BYTE	0–255
If the terminal does not support the specified resolution, it should capture at the closest available resolution and upload the captured image/video.			

- **Immediate capture command response message ID: 0x0805**

The format of the immediate capture command response message is shown in the table below. This message is in response to the immediate capture command 0x8801 delivered by the supervision platform.

Table 5-6 Format of Immediate Capture Command Response Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Reply sequence number	WORD	Correspond to the sequence number of the platform-initiated immediate capture command message (0x8801).
2	Result	BYTE	0: Success; 1: Failure; 2: Channel not supported The following fields are only valid if Result = 0.
3	ID count	WORD	-
4	ID list	BYTE[4*n]	n represents the number of captured multimedia items.

The platform sends: 7E 88 01 00 0C 12 02 46 96 05 19 00 05 02 00 01 00 02 01 04 01 80 40 40 40 99 7E

The terminal replies: 7E 08 05 00 0D 12 02 46 96 05 19 00 06 00 05 00 00 02 10 00 00 00 20 00 00 00 ED 7E

5.4 Multimedia Data Retrieval (0x8802 / 0x0802)

The platform sends a multimedia data retrieval message to obtain the status of stored multimedia data on the terminal. The terminal must reply with a multimedia data retrieval response message.

Based on the retrieval result, the platform can send a multimedia data upload message to request the terminal to upload the specified multimedia data. This message requires a general terminal response (0x0001).

- **Multimedia data retrieval message ID: 0x8802.**

The format for the multimedia data retrieval message body is shown in the table below. If no time range is specified, start time and end time are both set to 00-00-00-00-00-00.

Table 5-7 Format of Multimedia Data Retrieval Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Multimedia type	BYTE	0: Photo; 1: Audio; 2: Video
1	Channel ID	BYTE	0 means to retrieve all channels for data of the specified multimedia type.
2	Event item code	BYTE	0: Platform-issued command; 1: Timed action; 2: Burglary alarm; 3: Collision/rollover alarm; Others are reserved.
3	Start time	BCD[6]	YY-MM-DD-hh-mm-ss
9	End time	BCD[6]	YY-MM-DD-hh-mm-ss

- **Multimedia data retrieval response message ID: 0x0802**

The format of the multimedia data retrieval response message body is shown in the table below.

Table 5-8 Format of Multimedia Data Retrieval Response Message Body

<table>
<tr><th>Start Byte</th><th colspan="2">Field</th><th>Data Type</th><th>Description/Requirements</th></tr>
<tr><td>0</td><td colspan="2">Reply sequence number</td><td>WORD</td><td>Correspond to the sequence number of the multimedia data retrieval message.</td></tr>
<tr><td>2</td><td colspan="2">Total number of multimedia data items</td><td>WORD</td><td>Total number of multimedia data items (n) that meet the search criteria.</td></tr>
<tr><td rowspan="4">4</td><td rowspan="4">Retrieval item 1</td><td>Multimedia data ID</td><td>DWORD</td><td>Value greater than 0</td></tr>
<tr><td>Multimedia type</td><td>BYTE</td><td>0: Photo; 1: Audio; 2: Video</td></tr>
<tr><td>Channel ID</td><td>BYTE</td><td>-</td></tr>
<tr><td>Event item code</td><td>BYTE</td><td>0: Platform-issued command; 1: Timed</td></tr>
</table>

Start Byte	Field		Data Type	Description/Requirements
				action; 2: Burglary alarm; 3: Collision/rollover alarm; Others are reserved
		Location reporting (0x0200) message body	-	Indicate the location reporting message at the start of the capture or recording.
	Retrieval item 2	…		
	Retrieval item n	…		

The platform sends: 7E 88 02 00 0F 12 02 46 96 05 19 00 07 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 5E 7E

5.5 Single Stored Multimedia Data Retrieval and Upload (0x8805)

The format of the single stored multimedia data retrieval and upload message body is shown in the table below.

Table 5-9 Format of Single Stored Multimedia Data Retrieval and Upload Message Body

Start Byte	Field	Data Type	Description/Requirements
0	Multimedia data ID	DWORD	Value greater than 0
4	Delete flag	BYTE	0: Retain; 1: Delete

The platform sends: 7E 88 05 00 05 12 02 46 96 05 19 00 09 00 00 00 10 00 4D 7E

6. JT/T 1078-2016 Protocols (Message Body)

Table 6-1 Request-Response IDs Mapping Table

Event/Function	Request Message	Response Message
Query audio/video properties	[Platform] 0x9003 Terminal audio/video properties query	[Terminal] 0x1003 Terminal uploads audio/video properties
Query resource list	[Platform] 0x9205 Resource list query	[Terminal] 0x0001 General terminal response
		[Terminal] 0x1205 Terminal uploads audio/video resource list
File upload	[Platform] 0x9206 File upload command	[Terminal] 0x0001 General terminal response
	[Platform] 0x9207 File upload control	[Terminal] 0x0001 General terminal response
	[Terminal] 0x1206 File upload completion notification	[Platform] 0x8001 General platform response

6.1 Real-Time Audio and Video Transmission Command (0x9101)

Message type: Signaling data message

The platform initiates a request to the terminal for real-time audio/video data, including the transmission of real-time video, initiation of two-way voice communication, one-way listen-in, broadcasting audio to all terminals, and specific pass-through, etc. See the table below for the message body format. Upon receiving this message, the terminal replies with a general response message (0x0001). It then establishes a link using the corresponding server IP address and port number and sends the requested audio/video stream according to the audio/video stream transmission protocol.

Upon receiving a special alarm from the video terminal, the platform should proactively deliver this command to start real-time audio/video transmission without waiting for a manual confirmation.

The two-way communication function belongs to the 0x9101 protocol and the data type is 2 (two-way voice communication).

Table 6-2 Data Format of Real-Time Audio-Video Transmission Request

Start Byte	Field	Data Type	Description/Requirements
0	Server IP address length	BYTE	Length (n)
1	Server IP address	BYTE[21]	Real-time video server IP address
1+n	Server video channel listening port (TCP)	WORD	TCP port number of the real-time video server
3+n	Server video channel listening port (UDP)	WORD	UDP port number of the real-time video server
5+n	Logical channel number	BYTE	01–1F
6+n	Data Type	BYTE	0: Audio and video, 1: Video, 2: Two-way voice communication, 3: Listen-in, 4: Central broadcast, 5: Transparent transmission
7+n	Stream type	BYTE	0: Main stream; 1: Sub-stream

6.2 Live Audio-Video Streaming and Data Passthrough

Message type: Stream data message

Real-time audio/video stream transmission follows the RTP protocol framework, utilizing UDP or TCP as transport protocols. The payload packet format extends the IETF RFC 3550 RTP specification by incorporating additional fields such as message sequence number, SIM card number, and audio/video channel identifier. The payload structure is defined in the table below (populated in big-endian byte order).

Start Byte	Field	Data Type	Description & Requirements
0	Header Identifier	DWORD	Fixed value: 0x30 0x31 0x63 0x64
4	V	2 BITS	Fixed value: 2
	P	1 BIT	Fixed value: 0
	X	1 BIT	RTP header extension flag, fixed value: 0
	CC	4 BITS	Fixed value: 1
5	M	1 BIT	Marker bit: Indicates boundary of a complete data frame
	PT	7 BITS	Payload type (see the table below for details).
6	Sequence Number	WORD	Initialized at 0; increments by 1 per transmitted RTP packet
8	SIM Card Number	BCD[6]	Device ID
14	Logical Channel ID	BYTE	01–1F
15	Data Type	4 BITS	0000: Video I-frame

Start Byte	Field	Data Type	Description & Requirements
			0001: Video P-frame 0010: Video B-frame 0011: Audio frame 0100: Passthrough data
	Subpacket Handling Flag	4 BITS	0000: Atomic packet (indivisible) 0001: First packet 0010: Last packet 0011: Middle packet
16	Timestamp	BYTE[8]	Relative timestamp of current frame (unit: ms). Omitted for pass-through data (Type 0100)
24	Last Keyframe Interval	WORD	Time interval from previous video I-frame (unit: ms). Omitted for non-video frames.
26	Last Frame Interval	WORD	Time interval from previous frame (unit: ms). Omitted for non-video frames.
28	Data Body Length	WORD	Length of subsequent data body (excluding from this data packet)
30	Data Body	BYTE[n]	Audio/video or pass-through data. Maximum length: 950 bytes.

Table 6-3 Definitions of Audio-Video Coding Formats

Code	Name	Remarks
0	Reserved	
1	G. 721	Audio
2	G. 722	Audio
3	G. 723	Audio
4	G. 728	Audio
5	G. 729	Audio
6	G. 711A	Audio
7	G. 711U	Audio
8	G. 726	Audio
9	G. 729A	Audio
10	DVI4_3	Audio
11	DVI4_4	Audio
12	DVI4_8K	Audio
13	DVI4_16K	Audio

Code	Name	Remarks
14	LPC	Audio
15	S16BE_STEREO	Audio
16	S16BE_MONO	Audio
17	MPEGAUDIO	Audio
18	LPCM	Audio
19	AAC	Audio
20	WMA9STD	Audio
21	HEAAC	Audio
22	PCM_VOICE	Audio
23	PCM_AUDIO	Audio
24	AACLC	Audio
25	MP3	Audio
26	ADPCMA	Audio
27	MP4 AUDIO	Audio
28	AMR	Audio
29-90	Reserved	
91	Transparently transmitted	System
92-97	Reserved	Video
98	H.264	Video
99	H.265	Video
100	AVS	Video
101	SVAC	Video
102-110		Reserved
111-127		Custom

6.3 Real-time Audio and Video Transmission Control (0x9102)

Message type: Signaling data message

The platform sends real-time audio/video transmission control commands to switch the stream type, pause stream transmission, close transmission channels, etc. See the table below for the message body data format.

Table 6-4 Data Format of Real-Time Audio-Video Transmission Control Message

Start Byte	Field	Data Type	Description/Requirements
0	Logical channel number	BYTE	01–1F
1	Control command	BYTE	This command is used by the platform to control the transmission of the real-time audio/video data of the terminal. 0: Close audio/video transmission; 1: Switch stream (includes pause and resume); 2: Pause the transmission of all streams on this channel; 3: Resume the transmission of streams, maintaining the same stream type as before; 4: Close two-way voice communication.
2	Close audio/video type	BYTE	0: Close both audio and video data for this channel; 1: Close only audio, keep video; 2: Close only video, keep audio.
3	Switch stream type	BYTE	Switch the previously requested stream to the newly requested stream, with audio remaining the same stream as before. The new stream is: 0: Main stream; 1: Sub-stream.

6.4 Resource List Query (0x9205)

Message type: Signaling data message

The platform searches for the list of recording files in the terminal based on a combination of search criteria such as audio/video type, channel number, alarm type, and start/end time. See the table below for the message body format.

Table 6-5 Data Format of Recording File List Query Message

Start Byte	Field	Data Type	Description/Requirements
0	Logical channel number	BYTE	01–1F, where "0" indicates all channels.
1	Start time	BCD[6]	YY-MM-DD-HH-MM-SS, where all 0s indicate no start time is set.
7	End time	BCD[6]	YY-MM-DD-HH-MM-SS, where all 0s indicate no end time is set.
13	Alarm flag	64 BITS	For bits 0–31, see Table 4-22 Definitions of Alarm Flags for details; For bits 32–63, see Chapter 7 for details. All 0s indicate no alarm type is set.
21	Audio/video resource type	BYTE	0: Audio and video, 1: Audio, 2: Video, 3: Audio or video
22	Stream type	BYTE	0: All streams, 1: Main stream, 2: Sub-stream.
23	Memory type	BYTE	0: All memories, 1: Primary memory, 2: Backup memory

Table 6-6 Definitions of Video Alarm Flags

Bit	Definition	Processing Remarks
0	Video signal loss alarm	Flag persists until condition cleared.
1	Video signal block alarm	Flag persists until condition cleared.
2	Memory failure alarm	Flag persists until condition cleared.
3	Other video device failure alarm	Flag persists until condition cleared.
4	Passenger vehicle overcrowded alarm	Flag persists until condition cleared.
5	Abnormal driving behavior alarm	Flag persists until condition cleared.
6	Alarm about total size of special alarm videos reaching storage threshold	Zero upon receiving a response.
7–31	Reserved	

6.5 Audio and Video Resource List Upload (0x1205)

Message type: Signaling data message

This message is used by the terminal as a response to the platform's resource list query. If the list is too large and requires to be transmitted as segments, the segmentation will follow what is defined in Section 3.2. The platform should reply to each individual packet with a general response message (0x8001). See the table below for the message body format.

Table 6-7 Data Format of Audio-Video Resource List Upload Command Message

Start Byte	Field	Data Type	Description/Requirements
0	Sequence number	WORD	Correspond to the sequence number of the audio/video resource list query command
2	Total number of audio/video resources	DWORD	The number of audio-video resources meeting the criteria; set to 0 if there are none.
6	Audio/video resource list		See the table below.

Table 6-8 Format of Terminal-Upload Audio/Video Resource List

Start Byte	Field	Data Type	Description/Requirements
0	Logical channel number	BYTE	01–1F
1	Start time	BCD[6]	YY-MM-DD-HH-MM-SS
7	End time	BCD[6]	YY-MM-DD-HH-MM-SS
13	Alarm flag	64 BITS	For bits 0–31, see Table 4-22 Definitions of Alarm Flags for details; For bits 32–63, see Chapter 7 for details.
21	Audio/video resource type	BYTE	0: Audio and video, 1: Audio, 2: Video
22	Stream type	BYTE	1: Main stream; 2: Sub-stream
23	Memory type	BYTE	1: Primary memory; 2: Backup memory
24	File size	DWORD	Size in bytes

6.6 Platform-Initiated Remote Video Playback Request (0x9201)

The platform initiates a request to the terminal for audio/video playback. The terminal should respond using the 0x1205 message (Audio and Video Resource List Upload) and the video data transmission must follow the RTP packet format defined in Table 6-4 in Chapter 0. See the table below for the message body format.

Table 6-9 Data Format of Platform-Initiated Remote Video Playback Request

Start Byte	Field	Data Type	Description/Requirements
0	Server IP address length	BYTE	Length (n)
1	Server IP address	BYTE[21]	IP address of the real-time audio-video server
1+n	Server audio-video channel listening port (TCP)	WORD	TCP port number of the real-time audio-video server, which is set to "0" when the transmission is not using TCP.
3+n	Server audio-video channel listening port (UDP)	WORD	UDP port number of the real-time audio-video server, which is set to "0" when the transmission is not using UDP.
5+n	Logical channel number	BYTE	01–1F
6+n	Audio/video type	BYTE	0: Audio and video, 1: Audio, 2: Video, 3: Audio or video
7+n	Stream type	BYTE	0: Main stream or sub-stream, 1: Main stream, 2: Sub-stream; Set this field to 0 if this channel only transmits audio.
8+n	Memory type	BYTE	0: Primary or backup memory, 1: Primary memory, 2: Backup memory
9+n	Playback mode	BYTE	0: Normal; 1: Fast forward; 2: Keyframe rewind; 3: Keyframe playback; 4: Single frame upload.
10+n	Fast forward/rewind speed	BYTE	For playback modes 1 and 2, this field is valid; otherwise, set it to 0. 0: Invalid; 1: 1x; 2: 2x.
10+n	Fast forward/rewind speed	BYTE	3: 4x; 4: 8x; 5: 16x.
11+n	Start time	BCD[6]	YY-MM-DD-HH-MM-SS; This field indicates the upload time of a single frame when the playback

Start Byte	Field	Data Type	Description/Requirements
			mode is set to 4.
17+n	End time	BCD[6]	YY-MM-DD-HH-MM-SS; if it is set to 0, the playback continues indefinitely; this field is invalid when the playback mode is set to 4.

6.7 File Upload Command (0x9206)

Message type: Signaling data message

The platform issues a file upload command to the terminal. After receiving the command, the terminal responds with a general response message (0x0001) and uploads the file to the designated path on the target FTP server. See the table below for the message body format.

Table 6-10 Data Format of File Upload Command Message

Start Byte	Field	Data Type	Description/Requirements
0	Server address length	BYTE	Length (k)
1	Server address	BYTE[21]	FTP server address
1+k	Port	WORD	FTP server port number
3+k	Length of the username	BYTE	Length "l"
4+k	User name	BYTE[21]	User name for the FTP server
4+k+l	Password length	BYTE	Length (m)
5+k+l	Password	BYTE[21]	Password for the FTP server
5+k+l+m	File upload path length	BYTE	Length (n)
6+k+l+m	File upload path	BYTE[21]	File upload path
6+k+l+m+n	Logical channel number	BYTE	01–1F
7+k+l+m+n	Start time	BCD[6]	YY-MM-DD-HH-MM-SS
13+k+l+m+n	End time	BCD[6]	YY-MM-DD-HH-MM-SS
19+k+l+m+n	Alarm flag	64 BITS	For bits 0–31, see Table 4-22 Definitions of Alarm Flags for details; For bits 32–63, see Chapter 7 for details. All 0s indicate no alarm type is set.
27+k+l+m+n	Audio/video resource type	BYTE	0: Audio and video, 1: Audio, 2: Video, 3: Audio or video
28+k+l+m+n	Stream type	BYTE	0: All streams, 1: Main stream, 2: Sub-stream.
29+k+l+m+n	Store location	BYTE	0: Primary or backup memory, 1: Primary

Start Byte	Field	Data Type	Description/Requirements
			memory, 2: Backup memory
30+k+l+m+n	Task execution condition	BYTE	Represent by bits: bit 0: WiFi (1: Downloadable via WiFi); bit 1: LAN (1: Downloadable via LAN); bit 2: 3G/4G (1: Downloadable via 3G/4G).

6.8 File Upload Completion Notification (0x1206)

Message type: Signaling data message

When all files are uploaded via FTP, the terminal sends this command to notify the platform. See the table below for the message body format.

Table 6-11 Data Format of File Upload Completion Notification Message

Start Byte	Field	Data Type	Description/Requirements
0	Reply sequence number	WORD	Correspond to the sequence number of the platform-initiated file upload command.
2	Result	BYTE	0: Success; 1: Failure.

6.9 File Upload Control (0x9207)

Message type: Signaling data message

The platform notifies the terminal to pause, resume, or cancel all files in transmission. See the table below for the message body format.

Table 6-12 Data Format of File Upload Control Message

Start Byte	Field	Data Type	Description/Requirements
0	Reply sequence number	WORD	Correspond to the sequence number of the platform-initiated file upload command.
2	Upload control	BYTE	0: Pause, 1: Resume, 2: Cancel

7. Appendix I: 0x0200 Supplementary Info IDs and Extension

Supplementary Information ID	Supplementary Information Length	Description/Requirements
0x01	4	Mileage (based on vehicle odometer reading), DWORD, unit: 1/10km
0x02	2	Fuel level (based on vehicle fuel gauge), WORD, unit: 1/10 L
……	......	......
0x14	4	Video-related alarm, DWORD, bitwise setting, see the table below for the definitions of flags.
0x15	4	Video signal loss alarm status, DWORD, bitwise setting, bits 0 to 31 correspond to logical channels 1 to 32. If the respective bit is 1, then video signal loss occurs on that channel.
0x17	2	Memory failure alarm status, WORD, bitwise setting. Bits 0 to 11 correspond to primary memories 1 to 12; while bits 12 to 15 correspond to backup memory equipment 1 to 4. If the respective bit is 1, then the failure occurs to that memory.
……	......	......
0x30	1	Radio network signal strength, BYTE
0x31	1	Number of positioning satellites, BYTE
……	......	......
0x64	X	ADAS (Advanced Driver Assistance System)
0x65	X	DMS (Driver Monitoring System)
......	......	......
0xE8	n	Custom extension
……	......	......
0xEB	n	Timezone extension
0xFF	n	Manufacturer custom extension

Device alarms are divided into two categories. These defined in the standard Ministry protocols are standard alarms, which are reported using standard supplementary information IDs. Other alarms are reported using the extended protocol 0xE8 in 0x0200 message.

Message ID: 0x0200.

The location reporting message body consists of the basic and supplementary location information, as shown in the table below:

Table 7-1 Location Reporting Message Structure

Basic location information	Supplementary information item list

Table 7-2 Supplementary Information Item Format

Field	Data Type	Description/Requirements
Supplementary information ID	BYTE	1–255
Supplementary information length	BYTE	
Supplementary information		See previous context for the definition.

7.1 Standard Alarms

These alarms are already defined in the standard Ministry protocols and are mainly represented in 0x64 and 0x65.

Table 7-3 Extended Definition Table for Supplementary Information

Supplementary Information ID	Supplementary Information Length	Description/Requirements
0x64		ADAS alert information, as defined in Section 7.1.1.
0x65		DMS alert information, as defined in Section 7.1.2.

7.1.1 Standard ADAS Alarms

This section primarily references T/JSATL12-2017.

Table 7-4 Data Format of ADAS Alert Message

Start Byte	Field	Data Type	Description/Requirements
0	Alert ID	DWORD	Start from 0 and increment based on alert generation time regardless of alert type.
4	Status flag	BYTE	0x00: Unavailable 0x01: Start flag 0x02: End flag This field is applicable only to alerts or events with start and end flags. For alerts/events without start/end flags, this field

Start Byte	Field	Data Type	Description/Requirements
			should be set to 0x00.
5	Alert / event type	BYTE	0x01: Forward Collision Warning (FCW) 0x02: Lane Departure Warning (LDW) 0x03: Headway Monitoring and Warning (HMW) 0x04: Pedestrian Collision Warning (PCW); 0x05: Frequent lane change alarm; 0x06: Speed over road-sign-limit; 0x07: Obstacle warning; 0x08–0x0F: User defined; 0x10: Road sign recognition event; 0x11: Active capture event; 0x12–0x1F: User defined.
6	Alert level	BYTE	0x01: L1 alert 0x02: L2 alert
7	Front vehicle speed	BYTE	Unit: km/h; Range: 0–250; This field is valid only when the alert type is 0x01 or 0x02.
8	Distance to front vehicle/pedestrian	BYTE	Unit: 100ms; Range: 0–100; This field is valid only when the alert type is 0x01, 0x02, or 0x04.
9	Lane departure type	BYTE	0x01: Left departure 0x02: Right departure This field is valid only when the alert type is 0x02.
10	Recognized road sign type	BYTE	0x01: Speed limit 0x02: Height limit 0x03: Weight limit This field is valid only when the alert type is 0x06 or 0x10.
11	Road sign recognition data	BYTE	Data about the recognized road sign
12	Vehicle speed	BYTE	Unit: km/h; Range: 0–250
13	Elevation	WORD	Altitude in meters (m)
15	Latitude	INT32	Latitude in degrees multiplied by 10^6, accurate to one millionth of a degree
19	Longitude	INT32	Longitude in degrees multiplied by 10^6, accurate to one millionth of a degree
23	Date/Time	BCD[6]	YY-MM-DD-hh-mm-ss (GTM+8)
29	Vehicle status	WORD	See Table 7-6 Real-Time Data Format for details.
31	Alert ID	BYTE[16]	See the table below for details.

Table 7-5 Alert ID Format

Start Byte	Field	Data Type	Description/Requirements
0	Terminal ID	BYTE[7]	Consist of upper-case letters and digits.
7	Date/time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT time)
13	Serial No.	BYTE	Sequence number for alarms occurred at the same timestamp, incrementing from 0.
14	Attachment count	BYTE	Number of attachments for this specific alert
15	Reserved	BYTE	

Table 7-6 Real-Time Data Format

Start Byte	Field	Data Type	Description/Requirements
0	Vehicle speed	BYTE	Unit: km/h; Range: 0–250
1	Reserved	BYTE	
2	Mileage	DWORD	Unit: 0.1km; Range: 0–99999999
6	Reserved	BYTE[2]	
8	Elevation	WORD	Altitude in meters (m)
10	Latitude	INT32	Latitude in degrees multiplied by 10^6, accurate to one millionth of a degree
14	Longitude	INT32	Longitude in degrees multiplied by 10^6, accurate to one millionth of a degree
18	Date/Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT time)
24	Vehicle status	WORD	Other vehicle status by bit: [0] ACC status: 0-OFF, 1-ON [1] Left turn signal: 0-OFF, 1-ON [2] Right turn signal: 0-OFF, 1-ON [3] Wiper status: 0-OFF, 1-ON [4] Braking status: 0-Not braking, 1-Braking [5] Card insertion status: 0-No card, 1-Card inserted Bits 6–9: Custom [10] Positioning status: 0-No position fix, 1-Position fix acquired Bits 11–15: Custom

Alert types 0x01, 0x02, 0x03, and 0x04 have already realized in the dashcam.

7.1.2 Standard DMS Alarms

This section primarily references T/JSATL12-2017 and T/GDRTA 002-2020.

Table 7-7 Data Format of DMS Alert Message

Start Byte	Field	Data Type	Description/Requirements
0	Alert ID	DWORD	Start from 0 and increment based on alert generation time regardless of alert type.
4	Status flag	BYTE	0x00: Unavailable 0x01: Start flag 0x02: End flag This field is applicable only to alerts or events with start and end flags. For alerts/events without start/end flags, this field should be set to 0x00.
5	Alarm / event type	BYTE	0x01: Fatigue driving; 0x02: Phone call; 0x03: Smoking; 0x04: Distracted driving (looking down or around); 0x05: Driver abnormality; 0x06: Camera blocked; 0x08: Over-driving; 0x0A: Seatbelt unfastened; 0x0B: Sunglasses detected; 0x0C: Hands off steering wheel; 0x0D: Phone use; 0x10: Active capture event; 0x11: Driver change event; 0x12–0x1F: User defined.
6	Alert level	BYTE	0x01: L1 alert 0x02: L2 alert
7	Fatigue level	BYTE	Range: 0–10 The larger the value, the more fatigue the driver. This field is valid only when the alert type is 0x01.
8	Reserved	BYTE[4]	Reserved
12	Vehicle speed threshold	BYTE	Unit: km/h; Range: 0–250
13	Elevation	WORD	Altitude in meters (m)
15	Latitude	INT32	Latitude in degrees multiplied by 10^6, accurate to one millionth of a degree
19	Longitude	INT32	Longitude in degrees multiplied by 10^6, accurate to one millionth

Start Byte	Field	Data Type	Description/Requirements
			of a degree
23	Date/Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT time)
29	Vehicle status	WORD	See Table 7-6 Real-Time Data Format for details.
31	Alert ID	BYTE[16]	See Table 7-5 Alert ID Format for details.

Types 0x01, 0x02, 0x03, 0x04, 0x05 (face lost), 0x06 (camera obstructed), and 0x0B (sunglasses detected) have already realized in the dashcam.

7.2 E8 Extended Alarms (Custom)

In addition to the standard alarms defined in the Ministry's standard protocol, we support custom alarms using the 0xE8 protocol. The supplementary data format for the 0xE8 extended protocol is shown below:

Table 7-8 E8 Extended Alarm Data Structure

Field		Data Type	Description/Requirements
Alarm ID/event ID		WORD	0x0000–0x03FF: Video-related events/alarms (classified via video analysis); 0x0400–0x0BFF: Location-related events/alarms (related to driving routes or durations); 0x0C00–0x0FFF: Basic information events/alarms.
Alarm/event ID length		BYTE	Length of the identifier (0 if no ID; otherwise, it must be 16 bytes).
Alarm/event ID	Terminal ID	BYTE[7]	Consist of upper-case letters and digits.
	Date/time	BCD[6]	YY-MM-DD hh-mm-ss (BCD encoded system time)
	Serial No.	BYTE	Sequence number for alarms occurred at the same timestamp. It must be unique and it is recommended to increment from 0 cyclically.
	Attachment count	BYTE	Number of attachments
	Reserved	BYTE	It is 1 byte long and the default value is 0.
Alarm/event supplementary information length		BYTE	Length of the supplementary information in bytes (0 if no supplementary information).
Alarm/event supplementary information	ID1	WORD	
	Content 1 length	BYTE	

Field		Data Type	Description/Requirements
	(bytes)		
	Content 1	BYTE[n]	
	ID2	WORD	
	Content 2 length (bytes)	BYTE	
	Content 2	BYTE[n]	

Alarm/event IDs range from 0x0000–0x0FFF, which are further divided into three categories:

Table 7-9 Extended Alarm ID Ranges

Alarm Category	Alarm ID Range	Description
Video alarms	0x0001–0x03FF	Camera or algorithm related
Location alarms	0x0400–0x0BFF	GNSS, vehicle, or driving behavior related
Basic alarms	0x0C00–0x0FFF	SOS button, peripheral, battery, or system related

7.2.1 Video Alarms

Alarms supported by the dashcam are listed below:

Table 7-10 Extended Video Alarm IDs

Alarm ID	Description
0x0001	Camera failure
0x0002	Camera obstructed
0x0003	Seatbelt unfastened (ANWSB)
0x0004	Seatbelt fastened (AWSB)
0x0005	Face ID failed (AFIF)
0x0006	Face ID succeeded (AFIS)
0x0007	Eyes closed
0x0008	Yawning
0x0009	Face alignment failed
0x000A	Face lost
0x000B	Drinking
0x000C	Driver changed

7.2.2 Location/Driving Alarms

Table 7-11 Extended Location/Driving Alarm IDs

Alarm ID	Description
0x0400	Harsh acceleration
0x0401	Harsh braking
0x0402	Sharp cornering
0x0403	Overspeed
0x0404	Excessive driving time
0x0405	Driving collision
0x0406	Parking vibration
0x0407	Towing
0x0408	Geofence entry
0x0409	Geofence exit
0x040A	Left turn alarm
0x040B	Right turn alarm
0x040C	Door open alarm
0x040D	Door close alarm
0x0410	Sleep mode event
0x0411	Working mode event
0x0412	UBI harsh acceleration
0x0413	UBI harsh braking
0x0414	UBI sharp cornering
0x0415	UBI sudden lane change
0x0416	UBI collision
0x0417	UBI rollover
0x0418	UBI abnormal attitude
0x0419	UBI abnormal Euler

Example message for alarms 0x0412 and 0x0410:

7e 02 00 00 a0 4e b6 fb 4a d6 06 02 49 00 00 08 00 00 4c 00 03 01 58 9f 53 06 ca 4e 41 00 00 01 1a 01 23 24 09 03 15 30 32 01 04 00 00 00 00 25 04 00 00 00 00 30 01 1f 31 01 1f eb 09 55 54 43 2b 30 38 3a 30 30 14 04 00 00 00 01 15 04 00 00 00 04 65 2f 00 00 00 77 00 ff 02 00 00 00 00 00 1c 00 00 01 58 9f 53 06 ca 4e 41 24 09 03 15 30 32 04 00 00 00 00 00 00 00 24 09 03 15 30 32 77 02 00 e8 28 **04 12** 10 00 00 00 00 00 00 00 24 09 03 15 30 29 76 06 00 00 00 **04 10** 00 00 00 00 00 00 00 24 09 03 15 30 32 77 06 00 00 ca 7e

7.2.3 Basic Alarms

Table 7-12 Extended Basic Alarm IDs

Alarm ID	Description
0x0C01	SOS emergency alert
0x0C02	Low external battery alert
0x0C03	ACC ON
0x0C04	ACC OFF
0x0C05	Theft alarm
0x0C06	DMS calibration error
0x0C07	Identity recognition alert
0x0C08	Door alert
0x0C09	Fuel sensor abnormal alert
0x0C0A	Temperature/humidity abnormal alert
0x0C0B	Card login alert (DLT)
0x0C0C	Card logout alert (DLT)
0x0C0D	Unauthorized card alert (DLT)
0x0C0E	Power failure alert
0x0C0F	Low internal battery alert
0x0C10	Shutdown (due to low battery) alert
0x0C11	Ambient sound alert
0x0C12	Tamper alert
0x0C13	Active offline alert
0x0C14	SD/TF card inserted or mounted
0x0C15	SD/TF card not inserted or removed
0x0C16	SD/TF card write error
0x0C17	Data overage alert
0x0C1A	Failed to download the audio file from the specified HTTP URL.

7.3 Positioning Data (Custom Extension)

The platform requires location data with status and relevant information, including the reason for reporting. Since this functionality is not defined in standard Ministry protocols, the following is a Jimi IoT custom extension using the additional 0information ID 0xE8 in 0x0200 message, where a 2-byte identifier is added in the 0xE8 data.

7.3.1 Location Data Status (0x2002)

Field	Data Type	Description
Location data status packet ID	WORD	0x2002 (fixed)
Location data status packet length	BYTE	
Data upload mode	BYTE	0x00: Timed reporting 0x01: Distance-based reporting 0x02: Cornering-based reporting (upload when the device turns abruptly) 0x03: ACC status change reporting 0x1C: INS position fix reporting
Real-time or buffered data upload status	BYTE	0x00: Real-time upload 0x01: Buffered upload (after network recovery)

7.3.2 Terminal Status Information Reporting (0x2003)

Field	Data Type	Description
Location data status packet ID	WORD	0x2003 (fixed)
Location data status packet length	BYTE	7
Device charging status	BYTE	0x00: Not charging 0x01: Charging
Internal battery level	BYTE	0x00: Not charging 0x01: Power extremely low (insufficient to initiate calls or messages) 0x02: Power very low (low battery alert) 0x03: Power low (normal use is possible) 0x04: Power medium 0x05: Power high

Field	Data Type	Description
		0x06: Power extremely high 0xFF: Voltage detection unsupported
Battery voltage	WORD	Conversion method: Convert hex to decimal, then divide by 10. Take 0x01 0x9F for example, 019F is 415 in decimal and 4.15 after being divided by 100, the resulting value is 4.15V.
Cellular signal strength	BYTE	0x00: No signal 0x01: Signal extremely weak 0x02: Signal very weak 0x03: Signal good 0x04: Signal strong
External battery voltage	WORD	Conversion method: Convert hex to decimal, then divide by 10 Take 0x01 0x9F for example, 019F is 415 in decimal and 4.15 after being divided by 100, the resulting value is 4.15V.

7.3.3 Terminal Status Information Reporting (0x2005)

Field		Data Type	Description
Terminal status packet ID		WORD	0x2005 (fixed)
Terminal status packet length		BYTE	If the packet length exceeds 255 bytes, a new ID should be defined.
Status information	ID1	WORD	Status information
	Length	BYTE	
	Content	BYTE[n]	
	ID2	WORD	
	....	....	

(1) Device status IDs (0x0000–0x1FFF)

Status Type	Status ID	Data Type	Description
Internal battery voltage	0x0001	WORD	Convert hex to decimal, then divide by 100 to get the voltage value. Unit: V. E.g.: 0x019F is 415 in decimal and 4.15 after being divided by 100.
Daily data usage	0x0002	DWORD	Report once per day and reset every month. Unit: KB

Status Type	Status ID	Data Type	Description
Internal battery level	0x0003	BYTE	Unit: % (percentage); Range: 0–100
Charging status	0x0004	BYTE	0x00: Charging; 0x01: Discharging
SIM card ICCID	0x0005	BYTE[10]	Hexadecimal. Integrated Circuit Card Identifiers (ICCIDs) are typically 19 or 20 digits, with leading zeros for 19-digit ICCIDs. The format may include: First 6 digits: carrier code (e.g., 898600 or 898602 for China Mobile); Middle part: area code, card type, etc.; Last 1-2 digits: checksum.
Network type	0x0006	BYTE	0: No network; 1: Wi-Fi; 2: 2G (GSM); 3: 3G (WCDMA); 4: 4G (LTE).
GPS HDOP	0x0007	WORD	The Horizontal Dilution of Precision (HDOP) value from the NMEA data stream (GNGGA or GNGSA sentence). To calculate the HDOP value, convert the hexadecimal value to decimal and divide by 10, e.g., 0x000F=1.5.

(2) Device peripheral status IDs (0x2000–0x1FFF)

Status Type	Status ID	Data Type	Description
External battery voltage	0x2001	WORD	Convert hex to decimal, then divide by 10 to get the voltage value. Unit: V.

7.3.4 Peripheral Raw Data Pass-Through (0x2007)

Field		Data Type	Detailed explanation
Peripheral information reporting packet ID		WORD	0x2007 (fixed)
Peripheral ID information reporting packet Length		BYTE	If the Length is less than 255, it is 1 byte; if the Length is greater than 255, it is 2 bytes.
Peripheral information	ID	WORD	
	Length	BYTE	
	Content	BYTE[n]	
	ID1	WORD	
	Length 1	BYTE	

Field		Data Type	Detailed explanation
	Content 1	BYTE[n]	

7.3.5 Peripheral Information (0x2006)

Field		Data Type	Description
Peripheral Information Packet ID		WORD	Fixed value: 0x2006
Packet Length		BYTE	The total length of the packet, which must be no greater than 255 bytes; otherwise, a new packet ID is needed.
Peripheral Information	ID1	WORD	
	Length	BYTE	
	Content	BYTE[n]	
	ID2	WORD	
	....	....	

Peripheral IDs (0x0001-0x7FFF)

Category	ID	Data Type	Value	Content Description
Fuel Sensor Data	0x0001	BYTE[5]	[1] Type [1] Path [2] Value [1] Unit	Type 0x00: Ultrasonic fuel sensor 0x01: Capacitive fuel sensor Path Fuel tank number (default: 0x00) Value: The voltage value For type=0: the value is a positive integer accurate to one decimal place (e.g., 0x00 0x06 = 0.6). For type=1, the value is a positive integer accurate to two decimal places (e.g., 0x00 0x06 = 0.06). Unit: 0x01: Height (cm), e.g., 19.47cm. 0x02: Percentage 0x03: Voltage 0x04: Volume (liters)

Category	ID	Data Type	Value	Content Description
Temperature Sensor Data	0x0002	WORD	[2] Temperature value	The temperature value is converted from hexadecimal to decimal, accurate to one decimal place. The unit is 0.1 ℃. For example, 0x01 0x1B is 283 in decimal, resulting in a temperature of 28.3 ℃.
Temperature Sensor Raw Data	0x0003	BYTE[n]	[n] Serial port raw data	This is transparent data. For example, 32 38 2E 33 in hexadecimal converts to 28.3 ℃ in ASCII.
KD032 CAN Data	0x0004	BYTE[n]		This is the CAN data received via the serial port. For details, contact your sales to get the model-specific protocol.

7.4 EB Extended Alarm (Custom Timezone Information)

Field	Data Type	Description
Supplementary information ID	BYTE	0xEB (fixed)
Timezone string length	BYTE	
Timezone string	STRING	E.g.: UTC+08:00, UTC-08:00, Asia/Shanghai Note: Daylight Saving Time is supported.

8. Image/Video Upload Protocol (Image Server)

8.1 General

UTF-8 encoding default, and the data returned in JSON format

Context-Type: default is application/json charset=utf-8

Timestamp: UTC format, yyyy-MM-dd HH:mm:ss

Return data's attributes

Parameter	Types	Required	Description
code	string	Y	Result code
message	string	N	Code description
data	Object	N	Return data result

Transmission protocol

Data transmitted in JSON string format.

HTTP request Headers

Parameter	Types	Required	Description
Content - Type	String	Y	Fixed value "application/json; charset =UTF-8 "

Description of status code

Common status codes	
Code	Description
200	Request succeeded
400	Parameter error
500	System error

8.2 Agreement details

Upload interface

http://[IP address]/upload

HTTP request method

Support POST submission method. Since the uploaded file is different from other http request content, you must modify the Content-Type of the Header as follows:

Parameter	Types	Required	Description
Content-Type	String	Y	multipart/form-data

Request URL parameters

Parameter	Types	Required	Description
file	FilePart	Y	binary format
filename	String	Y	File name: must be unique, otherwise error happen during storage
timestamp	String	Y	Current time in milliseconds
sign	String	Y	Encrypted string, used to verify whether the request is legal. Encryption algorithm: MD5 encryption "filename + timestamp + secretKey" to be 32 bits lower letter, and then Base64 encryption of the encrypted string SecretKey: fixed value, jimidvr@123!443
callbackBody	String	N	Detail as below

callbackBody

Parameter	Require	Type	Description	Remark
businessType	YES	string	regularPicture remotePictureOrVideo eventAttachment historyVideo facePicture faceLibrary timeLapseVideo videoClip facePictureDownload	
imei	YES	string		
camera	YES	int		
shootType	NO	int	1= remote photo	

Parameter	Require	Type	Description	Remark
			2= remote video	
shootTime	NO	int	Video length	Unit is second
alarmTime	YES	long	Unix format	Unit is second
eventType	NO	string	Alert ID	
lat	YES	string	latitude	
lng	YES	string	longitude	
mimeType	YES	string	image/jpeg image/png video/mp4 video/h264 video/h265 video/ts audio/aac text/plain	
localFileName	YES	string		
videoBeginTime	NO	long	Unix format	Unit is millisecond
videoEndTime	NO	long	Unix format	Unit is millisecond
codeType	NO	int	1= main streaming 2= sub streaming	
timezone	NO	string		Example : "timezone":"GMT+03:00"
instructionId	NO	string	Unique identifier	

Response parameter

Parameter	Types	Required	Description
code	string	Y	Return result code, 200 success
message	string	N	Status code description information
data	String	N	Return file name

Example:

{
"code":200,
"data":"img07-1588139792287.png",
"message":"the high success"

}

{
"code":400,
"message":"File content is empty"
}

9. Appendix II: Uplink and Downlink Data Examples

1. 0x0100 Terminal Registration

Uplink (0x0100): 7e 01 00 00 25 4e b6 fb 4a d6 28 00 09 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 9a 7e

Downlink (0x8100): 7e 81 00 00 17 4e b6 fb 4a d6 28 00 09 00 09 00 4f 44 59 31 4e 44 63 34 4d 44 63 77 4d 44 41 78 4d 44 51 33 59 7e

2. 0x0102 Terminal Authentication

Uplink (0x0102): 7e 01 02 00 14 4e b6 fb 4a d6 28 00 0a 4f 44 59 31 4e 44 63 34 4d 44 63 77 4d 44 41 78 4d 44 51 33 d2 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 0a 00 0a 01 02 00 30 7e

3. 0x0002 Terminal Heartbeat

Uplink (0x0002): 7e 00 02 00 00 4e b6 fb 4a d6 28 00 0e bb 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 0e 00 0e 00 02 00 31 7e

4. 0x0200 Location Reporting

Uplink (0x0200): 7e 02 00 00 4b 4e b6 fb 4a d6 28 00 0b 00 00 08 00 00 0c 00 0d 00 00 00 00 00 00 00 00 00 00 00 00 00 00 25 01 21 17 13 09 01 04 00 00 00 00 25 04 00 00 00 00 30 01 16 31 01 00 eb 09 55 54 43 2b 30 38 3a 30 30 14 04 00 00 00 01 15 04 00 00 00 06 e8 04 04 11 00 00 81 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 0b 00 0b 02 00 00 31 7e

5. 0x0704 Batch Location Data Upload

Uplink (0x0704): 7e 07 04 01 95 4e b6 fb 4a d6 28 00 0c 00 06 01 00 39 00 00 00 00 00 0c 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 23 01 01 08 00 02 01 04 00 00 00 00 25 04 00 00 00 00 30 01 00 31 01 00 eb 09 55 54 43 2b 30 38 3a 30 30 00 39 00 00 00 00 00 0c 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 23 01 01 08 00 03 01 04 00 00 00 00 25 04 00 00 00 00 30 01 00 31 01 00 eb 09 55 54 43 2b 30 38 3a 30 30 00 45 00 00 08 00 00 0c 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 23 01 01 08 00 04 01 04 00 00 00 00 25 04 00 00 00 00 30 01 00 31 01 00 eb 09 55 54 43 2b 30 38 3a 30 30 14 04 00 00 00 01 15 04 00 00 00 06 00 45 00 00 08 00 00 0c 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 23 01 01 08 00 05 01 04 00 00 00 00 25 04 00 00 00 00 30 01 00 31 01 00 eb 09 55 54 43 2b 30 38 3a 30 30 14 04 00 00 00 01 15 04 00 00 00 06 00 45 00 00 08 00 00 0c 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 23 01 01 08 00 13 01 04 00 00 00 00 25 04 00 00 00 00 30 01 00 31 01 00 eb 09 55 54 43 2b 30 38 3a 30 30 14 04 00 00 00 01 15 04 00 00 00 06 00 45 00 00 08 00 00 0c 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 23 01 01 08 00 23 01 04 00 00 00 00 25 04 00 00 00 00 30 01 00 31 01 00 eb 09 55 54 43 2b 30 38 3a 30 30 14 04 00 00 00 01 15 04 00 00 00 06 1b 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 0c 00 0c 07 04 00 30 7e

6. 0x8900 Data Pass-Through

Downlink (0x8900): 7e 89 00 00 07 4e b6 fb 4a d6 28 dd b2 f0 73 65 72 76 65 72 a3 7e 55 54 43 2b 30 38 3a 30 30 14 04 00 00 00 01 15 04 00 00 00 06 e8 04 04 11 00 00 81 7e

Uplink (0x0001): 7e 00 01 00 05 4e b6 fb 4a d6 28 00 08 dd b2 89 00 00 5d 7e

Uplink (0x0900): 7e 09 00 00 3b 4e b6 fb 4a d6 28 dd b2 f0 73 65 72 76 65 72 2c 69 6f 74 68 75 62 2e 74 72 61 63 6b 73 6f 6c 69 64 70 72 6f 2e 63 6f 6d 2c 32 31 31 32 32 2c 31 31 33 2e 31 30 38 2e 36 32 2e 32 30 32 2c 35 32 35 37 35 1e 7e

7. 0x9101 Real-time Audio and Video Streaming (CH1)

Downlink (0x9101): 7e 91 01 00 17 4e b6 fb 4a d6 28 f3 d9 0f 31 32 34 2e 32 34 33 2e 31 38 34 2e 32 33 37 27 12 00 00 01 00 00 06 7e

Uplink (0x0001): 7e 00 01 00 05 4e b6 fb 4a d6 28 00 16 f3 d9 91 01 00 1f 7e

0x9101 Real-time Audio and Video Streaming (CH2)

Downlink (0x9101): 7e 91 01 00 17 4e b6 fb 4a d6 28 f3 da 0f 31 32 34 2e 32 34 33 2e 31 38 34 2e 32 33 37 27 12 00 00 02 00 00 06 7e

Uplink (0x0001): 7e 00 01 00 05 4e b6 fb 4a d6 28 00 17 f3 da 91 01 00 1d 7e

0x9101 Real-time Audio and Video Streaming (CH3)

Downlink (0x9101): 7e 91 01 00 17 4e b6 fb 4a d6 28 f3 db 0f 31 32 34 2e 32 34 33 2e 31 38 34 2e 32 33 37 27 12 00 00 03 00 00 06 7e

Uplink (0x0001): 7e 00 01 00 05 4e b6 fb 4a d6 28 00 18 f3 db 91 01 00 13 7e

8. 0x9205 Resource List Query

Downlink (0x9205): 7e 92 05 00 18 4e b6 fb 4a d6 28 f3 f8 01 25 01 23 00 00 00 25 01 23 23 59 59 00 00 00 00 00 00 00 00 00 00 11 7e

Uplink (0x1205): 7e 12 05 23 84 4e b6 fb 4a d6 28 00 20 00 03 00 01 f3 f8 00 00 00 52 01
25 01 23 00 00 57 25 01 23 00 03 57 00 00 00 00 00 00 00 00 00 00 68 00 00 01 25 01
23 00 03 57 25 01 23 00 06 57 00 00 00 00 00 00 00 00 00 00 00 00 90 00 00 01 25 01 23 00
06 57 25 01 23 00 09 58 00 00 00 00 00 00 00 00 00 00 00 00 90 00 00 01 25 01 23 00 09 58
25 01 23 00 10 43 00 00 00 00 00 00 00 00 00 00 00 00 24 00 00 01 25 01 23 00 22 54 25 01
23 00 25 55 00 00 00 00 00 00 00 00 00 00 00 00 68 00 00 01 25 01 23 00 25 55 25 01 23 00
28 12 00 00 00 00 00 00 00 00 00 00 00 00 70 00 00 01 25 01 23 00 37 23 25 01 23 00 40 24
00 00 00 00 00 00 00 00 00 00 00 00 68 00 00 01 25 01 23 00 40 24 25 01 23 00 42 42 00 00
00 00 00 00 00 00 00 00 00 00 70 00 00 01 25 01 23 00 51 32 25 01 23 00 54 33 00 00 00 00
00 00 00 00 00 00 00 00 70 00 00 01 25 01 23 00 54 33 25 01 23 00 56 51 00 00 00 00 00 00
00 00 00 00 00 00 94 00 00 01 25 01 23 01 06 20 25 01 23 01 09 21 00 00 00 00 00 00 00 00
00 00 00 00 88 00 00 01 25 01 23 01 09 21 25 01 23 01 11 38 00 00 00 00 00 00 00 00 00 00
00 00 74 00 00 01 25 01 23 01 20 28 25 01 23 01 23 29 00 00 00 00 00 00 00 00 00 00 00 00
64 00 00 01 25 01 23 01 23 29 25 01 23 01 25 47 00 00 00 00 00 00 00 00 00 00 00 00 6c 00
00 01 25 01 23 01 35 37 25 01 23 01 38 37 00 00 00 00 00 00 00 00 00 00 00 00 64 00 00 01
25 01 23 01 38 37 25 01 23 01 40 56 00 00 00 00 00 00 00 00 00 00 00 00 70 00 00 01 25 01
23 01 50 24 25 01 23 01 53 25 00 00 00 00 00 00 00 00 00 00 00 00 64 00 00 01 25 01 23 01
53 24 25 01 23 01 55 42 00 00 00 00 00 00 00 00 00 00 00 00 68 00 00 01 25 01 23 02 04 34
25 01 23 02 07 35 00 00 00 00 00 00 00 00 00 00 00 00 64 00 00 01 25 01 23 02 07 35 25 01
23 02 09 53 00 00 00 00 00 00 00 00 00 00 00 00 6c 00 00 01 25 01 23 02 19 01 25 01 23 02
22 02 00 00 00 00 00 00 00 00 00 00 00 00 68 00 00 01 25 01 23 02 22 02 25 01 23 02 24 20
00 00 00 00 00 00 00 00 00 00 00 00 6c 00 00 01 25 01 23 02 33 52 25 01 23 02 36 52 00 00
00 00 00 00 00 00 00 00 00 00 68 00 00 01 25 01 23 02 36 52 25 01 23 02 39 11 00 00 00 00
00 00 00 00 00 00 00 00 70 00 00 01 25 01 23 02 48 02 25 01 23 02 51 03 00 00 00 00 00 00
00 00 00 00 00 00 64 00 00 01 25 01 23 02 51 03 25 01 23 02 53 21 00 00 00 00 00 00 00 00
00 00 00 00 6c 00 00 01 25 01 23 03 01 59 25 01 23 03 04 59 00 00 00 00 00 00 00 00 00 00
00 00 64 00 00 01 25 01 23 03 04 59 25 01 23 03 07 18 00 00 00 00 00 00 00 00 00 00 00 00
70 00 00 01 25 01 23 03 16 04 25 01 23 03 19 04 00 00 00 00 00 00 00 00 00 00 00 00 68 00
00 01 25 01 23 03 19 04 25 01 23 03 21 23 00 00 00 00 00 00 00 00 00 00 00 00 70 00 00 01
25 01 23 03 30 55 25 01 23 03 33 56 00 00 00 00 00 00 00 00 00 00 00 00 68 00 00 01 25 01
23 03 33 56 25 01 23 03 36 14 00 00 00 00 00 00 00 00 00 00 00 00 70 0c 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 20 00 20 12 05 00 24 7e

9. 0x9201 Platform-Initiated Remote Video Playback Request

Downlink (0x9201): 7e 92 01 00 26 4e b6 fb 4a d6 28 f8 14 0f 31 32 34 2e 32 34 33 2e 31 38 34 2e 32 33 37 27 13 00 00 01 00 00 00 00 00 25 01 23 10 15 12 25 01 23 10 18 12 fe 7e

Uplink (0x1205): 7e 12 05 00 22 4e b6 fb 4a d6 28 00 13 f8 14 00 00 00 01 01 25 01 23 10 15 12 25 01 23 10 18 12 00 00 00 00 00 00 00 00 00 00 00 d8 00 00 a8 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 13 00 13 12 05 00 24 7e

10. 0x9206 File Upload Command

Downlink (0x9206): 7e 92 06 00 5c 4e b6 fb 4a d6 28 f8 3d 17 68 6b 66 74 70 2e 74 72 61 63 6b 73 6f 6c 69 64 70 72 6f 2e 63 6f 6d 00 15 07 6a 69 6d 69 66 74 70 0f 45 79 61 63 52 4d 69 73 4b 47 51 4e 51 55 65 10 2f 74 73 70 2f 63 34 35 30 2f 31 31 37 34 38 2f 01 25 01 23 07 21 50 25 01 23 07 24 09 00 00 00 00 00 00 00 00 00 00 01 b1 7e

Uplink (0x0001): 7e 00 01 00 05 4e b6 fb 4a d6 28 00 1b f8 3d 92 06 00 f9 7e

Uplink (0x1206): 7e 12 06 00 03 4e b6 fb 4a d6 28 00 1c f8 3d 00 79 7e

11. 0x8801 Remote Image Capture Comand

Downlink (0x8801): 7e 88 01 00 0c 4e b6 fb 4a d6 28 f8 6b 01 00 01 00 00 00 02 05 80 40 40 80 a6 7e

Uplink (0x0805): 7e 08 05 00 09 4e b6 fb 4a d6 28 00 21 f8 6b 00 00 01 07 91 be 9e b6 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 21 00 21 08 05 00 3e 7e

12. 0x0800 Multimedia Event Information Upload

Uplink (0x0800): 7e 08 00 00 08 4e b6 fb 4a d6 28 00 22 07 91 be 9e 00 00 00 01 22 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 22 00 22 08 00 00 3b 7e

13. 0x0801 Multimedia Data Upload

Uplink (0x0801): 7e 08 01 23 84 4e b6 fb 4a d6 28 00 23 00 6d 00 01 07 91 be 9e 00 00 00 01 00 00 00 00 00 0c 00 01 01 58 7d 01 ce 06 ca a2 10 00 00 00 00 00 25 01 23 11 59 26 ff d8 ff e0 00 10 4a 46 49 46 00 01 01 00 00 01 00 01 00 00 ff e1 00 e0 45 78 69 66 00 c3 4d 4d 00 2a 00 00 00 08 00 01 87 69 00 04 00 00 00 01 00 00 00 1a 00 00 00 00 00 0b 90 03 00 02 00 00 00 14 00 00 00 a4 a4 34 00 02 00 00 00 01 00 00 00 00 a4 35 00 02 00 00 00 01 00 00 00 00 92 02 00 05 00 00 00 01 00 00 00 b8 92 03 00 0a 00 00 00 01 00 00 00 c0 88 27 00 03 00 00 00 01 00 fc 00 00 82 9a 00 05 00 00 00 01 00 00 00 c8 92 04 00 0a 00 00 00 01 00 00 00 d0 88 22 00 03 00 00 00 01 00 02 00 00 a4 02 00 03 00 00 00 01 00 00 00 00 92 09 00 03 00 00 00 01 00 00 00 00 00 00 00 00 32 30 32 35 2d 30 31 2d 32 33 20 31 31 3a 35 39 3a 32 35 00 00 00 01 18 00 00 00 01 00 00 00 80 00 00 00 01 00 00 9c 3f 00 0f 42 40 00 00 00 00 00 00 00 01 ff db 00 43 00 08 0b 0c 0e 0c 0a 10 0e 0d 0e 12 11 10 13 18 29 1b 18 16 16 18 32 24 26 1e 29 3b 34 3e 3d 3a 34 39 38 41 49 5e 50 41 45 59 46 38 39 52 6f 53 59 61 64 69 6a 69 3f 4f 73 7b 72 66 7a 5e 67 69 65 ff db 00 43 01 08 12 12 18 15 18 30 1b 1b 30 65 43 39 43 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 65 ff c0 00 11 08 04 38 07 80 03 01 22 00 02 11 01 03 11 01 ff c4 00 1f 00 00 01 05 01 01 01 01 01 01 00 00 00 00 00 00 00 00 01 02 03 04 05 06 07 08 09 0a 0b ff c4 00 b5 10 00 02 01 03 03 02 04 03 05 05 04 04 00 00 01 7d 01 01 02 03 00 04 11 05 12 21 31 41 06 13 51 61 07 22 71 14 32 81 91 a1 08 23 42 b1 c1 15 52 d1 f0 24 33 62 72 82 09 0a 16 17 18 19 1a 25 26 27 28 29 2a 34 35 36 37 38 39 3a 43 44 45 46 47 48 49 4a 53 54 55 56 57 58 59 5a 63 64 65 66 67 68 69 6a 73 74 75 76 77 78 79 7a 83 84 85 86 87 88 89 8a 92 93 94 95 96 97 98 99 9a a2 a3 a4 a5 a6 a7 a8 a9 aa b2 b3 b4 b5 b6 b7 b8 b9 ba c2 c3 c4 c5 c6 c7 c8 c9 ca d2 d3 d4 d5 d6 d7 d8 d9 da e1 e2 e3 e4 e5 e6 e7 e8 e9 ea f1 f2 f3 f4 f5 f6 f7 f8 f9 fa ff c4 00 1f 01 00 03 01 01 01 01 01 01 01 01 01 00 00 00 00 00 00 01 02 03 04 05 06 07 08 09 0a 0b ff c4 00 b5 11 00 02 01 02 04 04 03 04 07 05 04 04 00 01 02 77 00 01 02 03 11 04 05 21 31 06 12 41 51 07 61 71 13 22 32 81 08 14 42 91 a1 b1 c1 09 23 33 52 f0 15 62 72 d1 0a 16 24 34 e1 25 f1 17 18 19 1a 26 27 28 29 2a 35 36 37 38 39 3a 43 44 45 46 47 48 49 4a 53 54 55 56 57 58 59 5a 63 64 65 66 67 68 69 6a 73 74 75 76 77 78 79 7a 82 83 84 85 86 87 88 89 8a 92 93 94 95 96 97 98 99 9a a2 a3 a4 a5 a6 a7 a8 a9 aa b2 b3 b4 b5 b6 b7 b8 b9 ba c2 c3 c4 c5 c6 c7 c8 c9 ca d2 d3 d4 d5 d6 d7 d8 d9 da e2 e3 e4 e5 e6 e7 e8 e9 ea f2 f3 f4 f5 f6 f7 f8 f9 fa ff da 00 0c 03 01 00 02 11 03 11 00 3f 00 f3 9a 28 a5 a0 41 4b 49 4b 40 c2 96 92 96 80 97 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 23 00 23 08 01 00 3a 7e

Downlink (0x8800): 7e 88 00 00 05 4e b6 fb 4a d6 28 00 01 07 a4 82 1a 00 00 7e

14. 0x8801 Remote Video Capture Command

Downlink (0x8801): 7e 88 01 00 0c 4e b6 fb 4a d6 28 fc 5f 01 ff ff 00 03 00 02 05 80 40 40 80 94 7e

Uplink (0x0805): 7e 08 05 00 09 4e b6 fb 4a d6 28 00 20 fc 5f 00 00 01 07 91 d7 2d 5d 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 20 00 20 08 05 00 3e 7e

15. 0x0200 Vibrating Alert Reporting (Alert ID: 0x0406)

Uplink (0x0200): 7e 02 00 00 4f 4e b6 fb 4a d6 28 00 08 00 00 00 00 00 4c 00 02 01 58 7d 01 e0 06 ca a2 4c 00 40 00 01 00 22 25 01 22 16 37 45 01 04 00 00 00 00 25 04 00 00 00 00 30 01 18 31 01 1c eb 09 55 54 43 2b 30 38 3a 30 30 e8 14 04 06 10 00 00 00 00 00 00 25 01 22 16 37 45 00 07 00 00 48 7e
Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 0b 00 0b 02 00 00 31 7e

Downlink (0x8900) – Alarm attachment upload request: 7e 89 00 00 58 4e b6 fb 4a d6 28 df 70 f0 56 49 44 45 4f 55 50 4c 4f 41 44 2c 68 6b 68 74 74 70 75 70 6c 6f 61 64 2e 74 72 61 63 6b 73 6f 6c 69 64 70 72 6f 2e 63 6f 6d 2c 34 34 33 2c 30 30 30 30 30 30 30 30 30 30 30 30 30 32 35 30 31 32 32 31 36 33 37 34 35 30 30 30 37 30 30 2c 31 2d 32 2d 33 2c 32 73 7e

Uplink (0x0001) – Terminal response: 7e 00 01 00 05 4e b6 fb 4a d6 28 00 09 df 70 89 00 00 9c 7e

Uplink (0x0900) – Upload start: 7e 09 00 00 13 4e b6 fb 4a d6 28 df 70 f0 73 74 61 72 74 20 75 70 6c 6f 61 64 20 74 61 73 6b 3b a7 7e

Uplink (0x0900) – CH1 image upload complete: 7e 09 00 00 4d 4e b6 fb 4a d6 28 df 70 f0 55 50 4c 4f 41 44 46 49 4c 45 43 4f 4d 50 4c 45 54 45 3a 38 36 35 34 37 38 30 37 30 30 30 31 30 34 37 5f 30 30 30 30 30 30 30 30 30 30 30 30 30 30 32 35 30 31 32 32 31 36 33 37 34 35 30 30 30 37 30 30 5f 31 5f 30 30 2e 6a 70 67 91 7e

Uplink (0x0900) – CH1 video upload complete: 7e 09 00 00 4d 4e b6 fb 4a d6 28 df 70 f0 55 50 4c 4f 41 44 46 49 4c 45 43 4f 4d 50 4c 45 54 45 3a 38 36 35 34 37 38 30 37 30 30 30 31 30 34 37 5f 30 30 30 30 30 30 30 30 30 30 30 30 30 30 32 35 30 31 32 32 31 36 33 37 34 35 30 30 30 37 30 30 5f 31 5f 30 30 2e 6d 70 34 c5 7e

Uplink (0x0900) – CH2 image upload complete: 7e 09 00 00 4d 4e b6 fb 4a d6 28 df 70 f0 55 50 4c 4f 41 44 46 49 4c 45 43 4f 4d 50 4c 45 54 45 3a 38 36 35 34 37 38 30 37 30 30 30 31 30 34 37 5f 30 30 30 30 30 30 30 30 30 30 30 30 30 30 32 35 30 31 32 32 31 36 33 37 34 35 30 30 30 37 30 30 5f 32 5f 30 30 2e 6a 70 67 92 7e

Uplink (0x0900) – CH2 video upload complete: 7e 09 00 00 4d 4e b6 fb 4a d6 28 df 70 f0 55 50 4c 4f 41 44 46 49 4c 45 43 4f 4d 50 4c 45 54 45 3a 38 36 35 34 37 38 30 37 30 30 30 31 30 34 37 5f 30 30 30 30 30 30 30 30 30 30 30 30 30 30 32 35 30 31 32 32 31 36 33 37 34 35 30 30 30 37 30 30 5f 32 5f 30 30 2e 6d 70 34 c6 7e

Uplink (0x0900) – CH3 image upload complete: 7e 09 00 00 4d 4e b6 fb 4a d6 28 df 70 f0 55 50 4c 4f 41 44 46 49 4c 45 43 4f 4d 50 4c 45 54 45 3a 38 36 35 34 37 38 30 37 30 30 30 31 30 34 37 5f 30 30 30 30 30 30 30 30 30 30 30 30 30 30 32 35 30 31 32 32 31 36 33 37 34 35 30 30 30 37 30 30 5f 33 5f 30 30 2e 6a 70 67 93 7e

Uplink (0x0900) – CH3 video upload complete: 7e 09 00 00 4d 4e b6 fb 4a d6 28 df 70 f0 55 50 4c 4f 41 44 46 49 4c 45 43 4f 4d 50 4c 45 54 45 3a 38 36 35 34 37 38 30 37 30 30 30 31 30 34 37 5f 30 30 30 30 30 30 30 30 30 30 30 30 30 30 32 35 30 31 32 32 31 36 33 37 34 35 30 30 30 37 30 30 5f 33 5f 30 30 2e 6d 70 34 c7 7e

16. 0x0200 SOS Alert Reporting (Alert ID: 0x0C01)

Uplink (0x0200): 7e 02 00 00 4f 4e b6 fb 4a d6 28 00 10 00 00 00 00 00 4c 00 03 01 58 7d 01 fb 06 ca a2 18 00 38 00 01 00 00 25 01 22 16 54 19 01 04 00 00 00 00 25 04 00 00 00 00 30 01 18 31 01 1e eb 09 55 54 43 2b 30 38 3a 30 30 e8 14 0c 01 10 00 00 00 00 00 00 00 25 01 22 16 54 19 01 07 00 00 48 7e

Downlink (0x8001): 7e 80 01 00 05 4e b6 fb 4a d6 28 00 10 00 10 02 00 00 31 7e

Downlink (0x8900) – Alarm attachment upload request: 7e 89 00 00 58 4e b6 fb 4a d6 28 e0 26 f0 56 49 44 45 4f 55 50 4c 4f 41 44 2c 68 6b 68 74 74 70 75 70 6c 6f 61 64 2e 74 72 61 63 6b 73 6f 6c 69 64 70 72 6f 2e 63 6f 6d 2c 34 34 33 2c 30 30 30 30 30 30 30 30 30 30 30 30 30 32 35 30 31 32 32 31 36 35 34 31 39 30 31 30 37 30 30 2c 31 2d 32 2d 33 2c 32 17 7e

Uplink (0x0001) – Terminal response: 7e 00 01 00 05 4e b6 fb 4a d6 28 00 11 e0 26 89 00 00 ed 7e

Uplink (0x0900) – Upload start: 7e 09 00 00 13 4e b6 fb 4a d6 28 e0 26 f0 73 74 61 72 74 20 75 70 6c 6f 61 64 20 74 61 73 6b 3b ce 7e

> **NOTE:** The attachment upload completion notification shares a common data structure with the vibrating alert. Due to space limitations, data examples for other event alerts (including DMS alerts, ADAS alerts, facial recognition alerts, overspeed alerts, UBI-related alerts, etc.) are not shown here.